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Chapter 2

Questions on Lesson Four

The Measuring Instruments

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Moving coil galvanometer

Current intensity

$$\text{Current intensity (I)} = \text{The equivalent of one division} \times \text{No. of divisions}$$

Galvanometer sensitivity

$$\text{Galvanometer sensitivity} = \frac{\theta}{I}$$

Moving coil ammeter

Ammeter resistance

$$R` = \frac{R_g R_s}{R_g + R_s}$$

Current intensity (I)

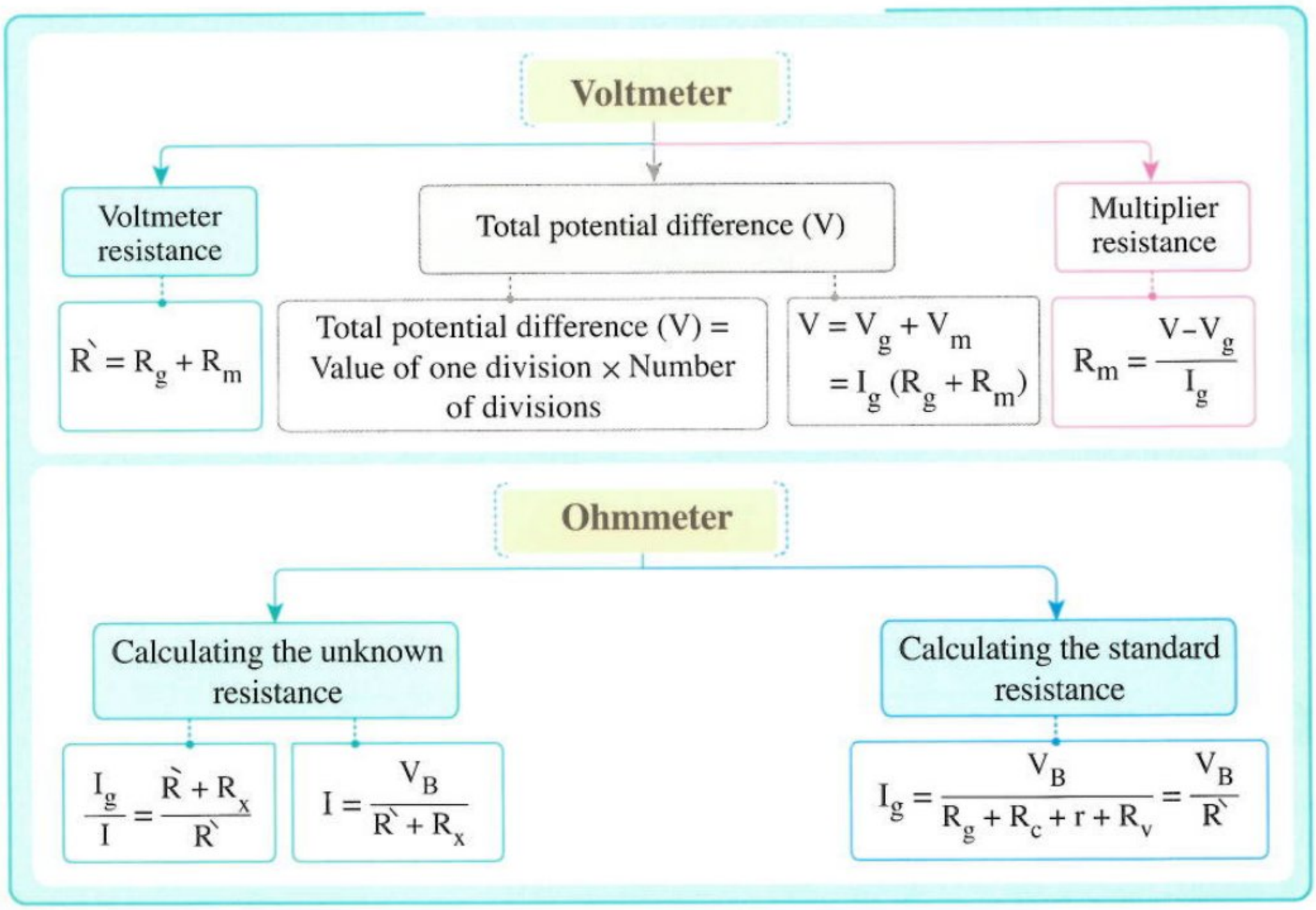
Current intensity (I) = Value of one division × Number of divisions indicated by the pointer

$$I = I_g + I_s = I_g + \frac{I_g R_g}{R_s}$$

Shunt resistance

$$R_s = \frac{I_g R_g}{I - I_g}$$

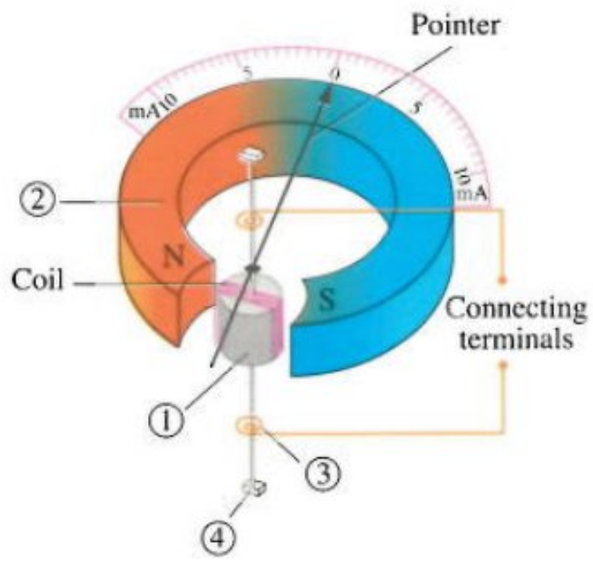
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First Multiple choice questions

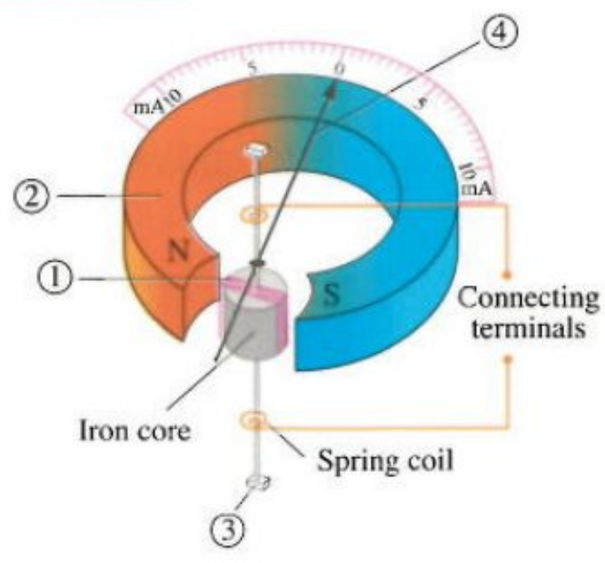
The moving coil galvanometer

1 The opposite figure represents the internal structure of the moving coil galvanometer, so the component which is made of agate to reduce friction of the coil during its motion is ..... .



(a) ①
(b) ②
(c) ③
(d) ④

2 The opposite figure represents the internal structure of the moving coil galvanometer, so the component which is made of copper is ..... .

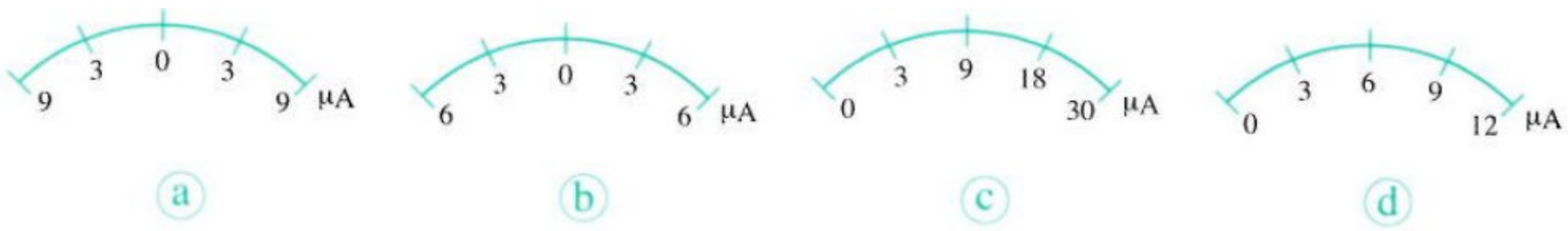


(a) ①
(b) ②
(c) ③
(d) ④

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3 Which of the following figures represents the measuring scale of a sensitive moving coil galvanometer to determine the direction of the electric current?



(a) (b) (c) (d)

4 The magnetic torque which acts on the coil of a galvanometer when an electric current passes through it always equals ..... .

(a) BIAN sin 0 (b) BIAN sin 45 (c) BIAN sin 90 (d) BIAN sin 30

5 The resultant torque acting on the coil of a galvanometer when its pointer stops at a certain reading equals ..... .

(a) BIAN (b) 2 BIAN (c) 2 BIAN sin θ (d) zero

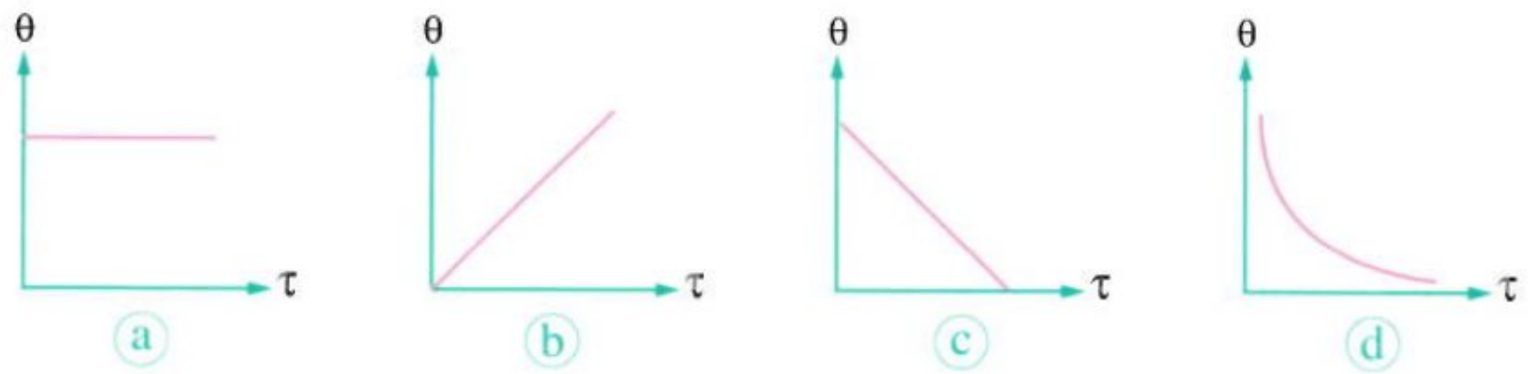
6 When a direct electric current passes through a coil of a sensitive galvanometer, its pointer takes a period t to deflect from zero position and settle at a certain reading, what happens to the values of each of the magnetic torque (τ_1) affected on the coil and the torsion (τ_2) for the two spring coils within that period?

(a) Increases, constant.
(b) Increases, increases.
(c) Constant, constant.
(d) Constant, increases.

7 When a direct electric current of intensity (I) passes through the coil of a galvanometer, the magnitude of the magnetic torque acting on the galvanometer coil does not change during its rotation due to ..... .

(a) the presence of spring coils
(b) the presence of agate bearings
(c) the presence of an aluminum pointer
(d) the concavity of the magnet poles and the presence of a soft iron core

8 ✱ Which of the following graphs represents the relation between the angle (θ) at which a galvanometer's pointer deflects from the zero position and the torque (τ) that affects the coil of the galvanometer due to the flow of a direct current?



(a) (b) (c) (d)

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9 When a suitable direct electric current passes in a moving coil galvanometer, what is the quantity that its value changes during the motion of the pointer starting from rest and till it stops at a certain reading?

(a) The magnetic torque acting on the coil.
(b) The torsion in the two spring coils.
(c) The sensitivity of the galvanometer.
(d) The angle between the coil and the direction of the magnetic flux.

10 When an alternating current of a very low frequency (1 Hz) passes through the galvanometer, the pointer of the galvanometer ..... .

(a) doesn't deflect from the zero point of its scale
(b) deflects and settles at a certain value
(c) vibrates about the sides of the zero point of its scale
(d) deflects to the end of its scale

11 When a DC electric current of high intensity passes through the coil of a galvanometer, then ..... .

(a) the galvanometer pointer does not deflect
(b) no torque is generated to act on the galvanometer coil
(c) high temperature is generated that may damage the coil
(d) the galvanometer sensitivity increases

12 If the pointer of the galvanometer has deflected by an angle of 50°, when a current of 500 μA passes through it, then the galvanometer sensitivity equals ..... deg/μA.

(a) 550 (b) 450 (c) 0.1 (d) 10

13 A galvanometer has a sensitivity of 2 deg/μA. Given that the value of each division is 5 μA and the maximum deflection angle of the galvanometer pointer from the zero position is 50°, so the number of divisions on the galvanometer scale is equal to ..... .

(a) 5 divisions (b) 10 divisions (c) 20 divisions (d) 25 divisions

14 A sensitive galvanometer scale is divided into 20 divisions. Its pointer deflects to half its scale when a current of 0.1 mA passes through its coil, so the indication of one division on the scale of the device equals ..... .

(a) 20 μA (b) 10 μA (c) 5 μA (d) 2 μA

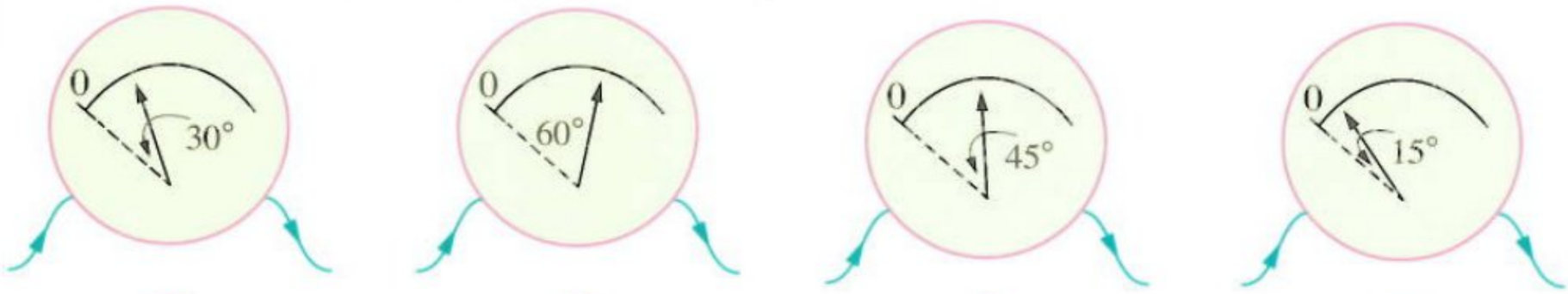
15 * A moving coil galvanometer shows half scale deflection when an electric current of 200 μA passes through it, hence if you know that each division denotes 0.08 mA, the number of divisions of the galvanometer's scale equals ..... .

(a) 10 divisions (b) 9 divisions (c) 7 divisions (d) 5 divisions

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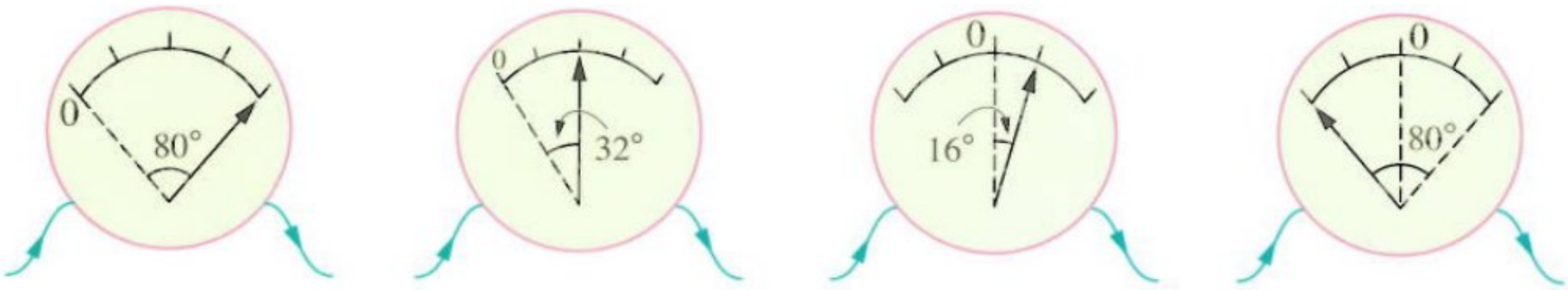
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16 The following figures represent the deflection in four sensitive galvanometers during the flow of the same electric current in each of them, which of them expresses the galvanometer with the higher sensitivity?



(a) (b) (c) (d)

17 The following figures represent the deflection of the pointer for four sensitive galvanometer devices when an electric current of 250 μA passes through each of them. Which of these devices has a sensitivity of 0.16 deg/μA?



(a) (b) (c) (d)

18 ✱ Two copper wires of the same length were used to manufacture the coils of two galvanometers P and Q, respectively. The wires were wound around square aluminum frame with side lengths X and 2 X, respectively. If both galvanometers have identical magnetic flux density affecting the coil and the same elasticity coefficient, then the ratio of $\frac{\text{the sensitivity of galvanometer P}}{\text{the sensitivity of galvanometer Q}}$ is equal to ..... .

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{2}{1}$ (d) $\frac{1}{1}$

19 ✱ A galvanometer coil that consists of 600 turns and has a cross-sectional area of 6 cm^2 is hung in a magnetic flux of density 0.1 T, hence the electric current intensity that is required to generate a torque of 4.32×10^{-3} N.m equals ..... .

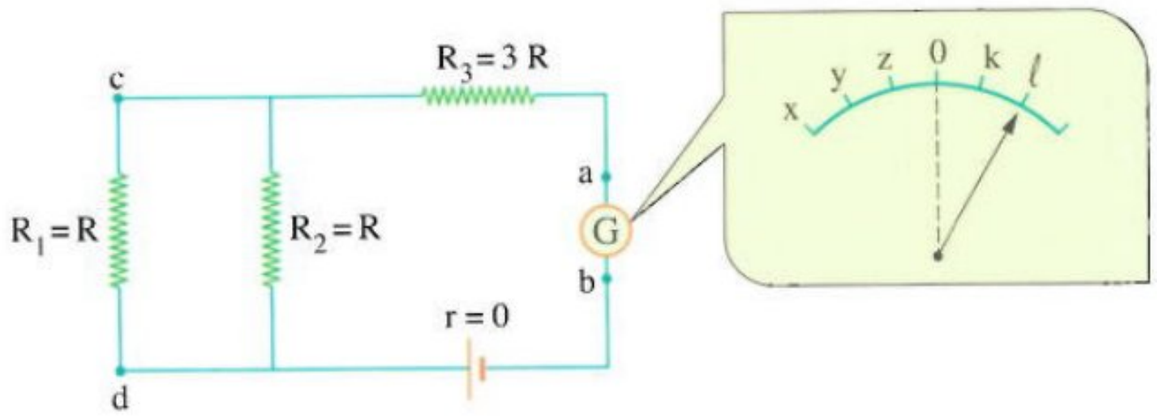
(a) 0.01 A (b) 0.02 A (c) 0.12 A (d) 0.21 A

20 ✱ A sensitive galvanometer has a coil that consists of 1200 turns where the face area of each turn is 3 cm^2. The coil rotates in a uniform magnetic field of flux density 0.01 T. When a current of intensity 1 mA passes in the coil of the galvanometer, its pointer gets deflected by an angle of 30° from the zero position, then the torsion of the two spring coils when the pointer of the galvanometer settles equals ..... .

(a) 2.55×10^{-6} N.m (b) 3.6×10^{-6} N.m
(c) 4.22×10^{-6} N.m (d) 6.4×10^{-6} N.m

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21 * A sensitive galvanometer with terminals a and b and its resistance R, with its zero position at the middle of its scale, its pointer deflects to position ℓ when connected in the electric circuit shown in the following diagram.



When the galvanometer's position in the circuit is swapped with resistor R_1, such that terminal a is connected to point c and terminal b is connected to point d, then the galvanometer pointer will deflect to position ..... .

(a) x (b) y (c) z (d) k

The ammeter

22 When the shunt resistance which is connected to the galvanometer decreases, the ammeter sensitivity ..... .

(a) increases (b) decreases
(c) remains unchanged (d) increases then decreases

23 The ratio between the shunt resistance to the total resistance of the ammeter is ..... one.

(a) greater than (b) equal to (c) less than (d) indeterminable

24 In the ammeter, when the shunt resistance which is connected to the galvanometer decreases while passing the same current, the quantity that increases is ..... .

(a) the torsion that affects the two spring coils
(b) the magnetic force which acts on the sides of the galvanometer coil
(c) the sensitivity of the ammeter
(d) the current measuring accuracy of the ammeter

25 An ammeter is connected in an electric circuit; hence it shows a full range deflection, what happens to the galvanometer if the shunt resistance is increased while the same current intensity flows in the ammeter?

(a) Its pointer deflects in the opposite direction.
(b) Its sensitivity decreases vastly.
(c) A current of intensity less than the end of the ammeter scale passes.
(d) The passing current in its coil will be greater than the maximum reading of its scale.

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26 In the ammeter, the range for measuring electric current intensity becomes greater when connecting the galvanometer with a shunt resistance of value that equals ..... .

(a) one third the galvanometer's resistance (b) double the galvanometer's resistance
(c) half the galvanometer's resistance (d) one fifth the galvanometer's resistance

27 A moving coil galvanometer shows full-scale deflection when a current (I) passes through it and when it is connected with a resistor of 12 Ω in parallel and allowing the same current (I) to pass, its pointer gets deflected to one fifth its scale, so the resistance of the galvanometer's coil (R_g) is ..... .

(a) 24 Ω (b) 36 Ω (c) 48 Ω (d) 60 Ω

28 If the ratio between the resistance of the ammeter and that of the galvanometer is $\frac{1}{10}$, then the ratio between the shunt resistance and the galvanometer resistance is ..... .

(a) $\frac{1}{10}$ (b) $\frac{10}{1}$ (c) $\frac{1}{99}$ (d) $\frac{1}{9}$

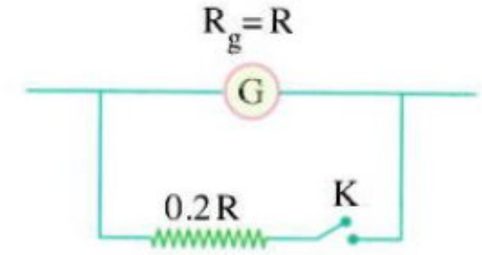
29 If the resistance of the coil of a galvanometer is R, so the shunt resistance that makes its sensitivity decrease to its quarter is ..... .

(a) R (b) $\frac{R}{2}$ (c) $\frac{R}{3}$ (d) $\frac{R}{4}$

30 ✱ A shunt resistance of 0.1 Ω reduces the sensitivity of an ammeter to the one-tenth of its initial value, so the shunt resistance that reduces the sensitivity of this ammeter to the quarter of its initial value equals ..... .

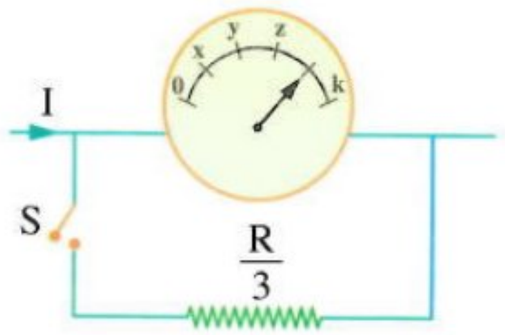
(a) 0.2 Ω (b) 0.3 Ω (c) 0.6 Ω (d) 0.9 Ω

31 In the opposite figure, a shunt resistance that is connected to a galvanometer through a switch K, when the switch is closed, the device sensitivity decreases to ..... its value.



(a) half (b) fifth
(c) sixth (d) quarter

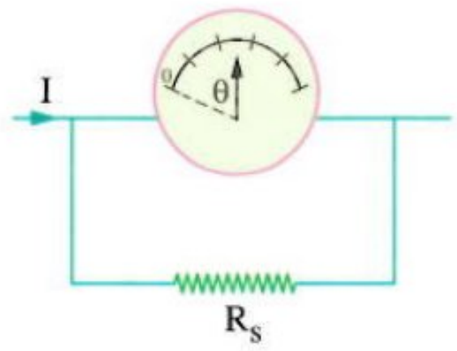
32 A galvanometer of resistance R is connected to a shunt resistance of value $\frac{R}{3}$ via a switch S to form an ammeter. When a current of intensity I passes through the galvanometer, its pointer deflects as shown in the opposite diagram. If switch S is closed and the same current passes through the ammeter, the galvanometer pointer will deflect to the position ..... .



(a) x (b) y (c) z (d) k

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33 An ammeter consists of a galvanometer and a shunt of resistance that equals the galvanometer resistance. When a current of intensity I passes in the ammeter, the pointer of the galvanometer deflects by an angle θ as in the opposite figure. If the shunt resistance is decreased to be one third that of the galvanometer while passing the same current (I) in the ammeter, then the deflection angle of the pointer from the zero position becomes ..... .



(a) θ (b) $\frac{\theta}{2}$ (c) $\frac{\theta}{3}$ (d) $\frac{\theta}{4}$

34 An ammeter that contains a galvanometer of resistance R_g is connected in a closed electric circuit, so if 2 % of the circuit current is passing through the galvanometer, then the shunt resistance equals ..... .

(a) $\frac{R_g}{50}$ (b) $\frac{R_g}{49}$ (c) 49 R_g (d) 50 R_g

35 * A galvanometer has a resistance of 54 Ω, if it is connected to a shunt resistance R_s, an electric current that equals 0.1 of the total current passes through its coil, so the value of resistance R_s equals ..... .

(a) 4.5 Ω (b) 6 Ω (c) 8.25 Ω (d) 10 Ω

36 A galvanometer's coil of resistance 36 Ω is connected to a shunt resistance of 4 Ω, then the resulted device is connected in a closed electric circuit, so the percentage of the current which passes in the galvanometer's coil to the total current equals ..... .

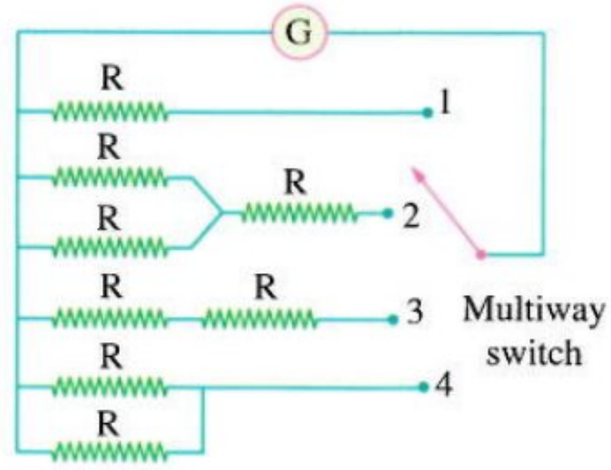
(a) 8% (b) 9% (c) 10% (d) 91%

37 * A galvanometer that has a coil of 0.1 Ω resistance shows full range deflection for a current of intensity I_g, hence the value of the shunt resistance that is required to increase the range of the galvanometer by 10 times as its initial range equals ..... .

(a) 0.03 Ω (b) 0.02 Ω (c) 0.01 Ω (d) 0.003 Ω

38 A sensitive galvanometer is connected to a multiway switch that can connect a galvanometer to one of the positions (1, 2, 3, 4) to convert it into ammeter, so the maximum measuring range of the ammeter can be achieved when connecting the switch at position ..... .

(a) 1 (b) 2
(c) 3 (d) 4



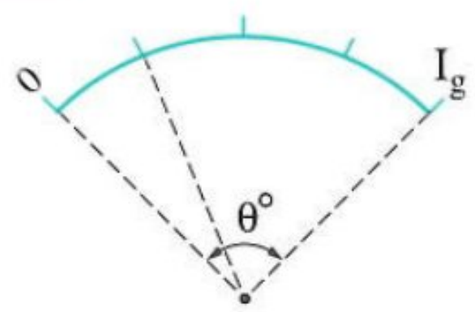
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39 ✱ A galvanometer of 54 Ω resistance shows a full-scale deflection for a current of 0.01 A. It is desired to be modified to measure a current of intensity 0.1 A, so ..... .

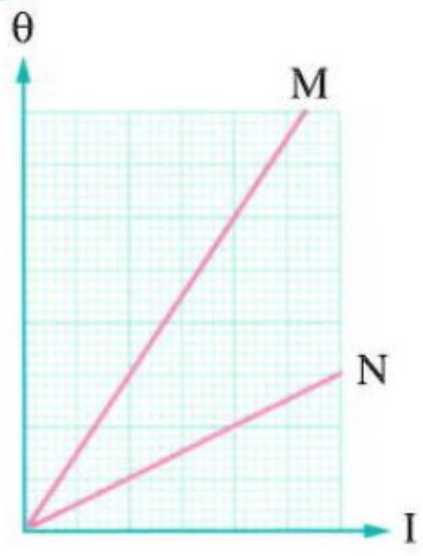
	The value of the shunt resistance	The method of connecting the resistance with the galvanometer
a	6 Ω	in series
b	3 Ω	in series
c	6 Ω	in parallel
d	3 Ω	in parallel

40 In the opposite figure , when an electric current of intensity (I_g) passes through a moving coil galvanometer, its pointer deflects by angle (θ) to the end of the scale and when it is connected to a shunt resistance to convert it into ammeter and the same current (I_g) passes through the ammeter, so the pointer deflects by an angle less than θ with ratio 75%, then the sensitivity of the device has ..... .



(a) increased by 75%
(b) decreased by 75%
(c) increased by 25%
(d) decreased by 25%

41 A galvanometer is connected to an electric circuit which its components are connected in series, the two graphical lines N, M represent the relation between the angle of deflection (θ) of the galvanometer's pointer and the current intensity (I) through the circuit before and after converting it into ammeter as shown in the opposite figure, then the sensitivity of the ammeter equals ..... .



(a) half the sensitivity of the galvanometer
(b) one third the sensitivity of the galvanometer
(c) quarter the sensitivity of the galvanometer
(d) the sensitivity of the galvanometer

42 A galvanometer was connected to a shunt resistance R_1, the maximum current the device could measure was (I) as in figure (1), when the same galvanometer was connected to a shunt resistance (R_2), the maximum current the device could measure was (2 I) as in figure (2). In which of the figures is the voltage difference across the terminals of the device greater?

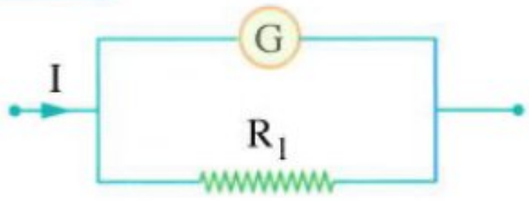


Figure (1)

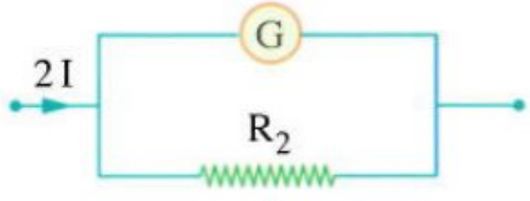
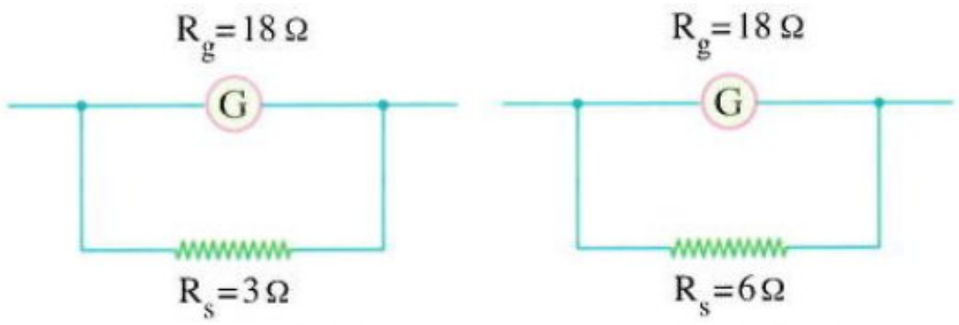


Figure (2)

(a) In figure (1)
(b) In figure (2)
(c) Both figures (1), (2).
(d) Indeterminable.

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43 The opposite figure shows two identical galvanometers. If each of them is connected to a shunt resistance to be converted into an ammeter, the ratio of the maximum currents that can be measured by both ammeters $(\frac{I_x}{I_y})$ is ..... .



(a) $\frac{4}{7}$ (b) $\frac{2}{1}$ (c) $\frac{3}{1}$ (d) $\frac{7}{4}$

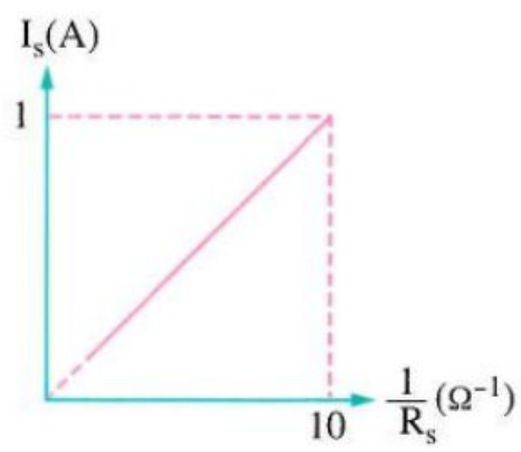
44 ✱ A moving coil galvanometer shows a full-scale deflection of 500 μA when a potential difference of 0.04 V is applied between its terminals, so the value of a shunt resistance, that can be used to convert it into an ammeter that can measure electric currents intensities up to a maximum value of 500 mA, equals ..... .

(a) 0.07 Ω (b) 0.035 Ω (c) 0.4 Ω (d) 0.08 Ω

45 ✱ An ammeter shows full-scale deflection when an electric current of 200 mA passes through it, also it shows a reading of 50 mA when a potential difference of 0.04 V is applied between its two terminals, so the value of a shunt resistance that enables the ammeter to measure electric currents up to a maximum value of 2 A equals ..... .

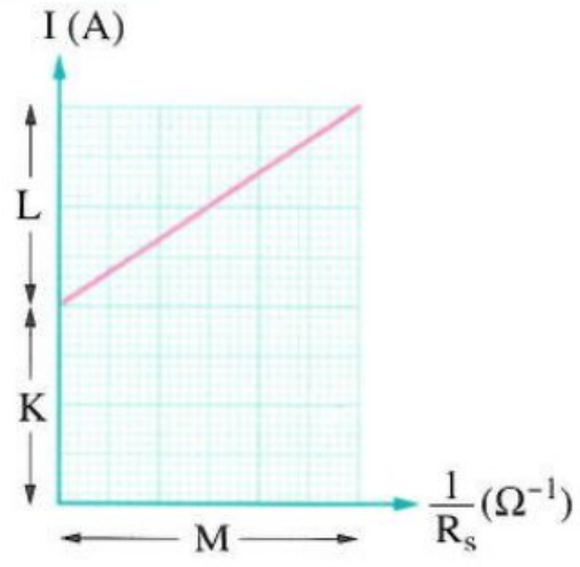
(a) 0.089 Ω (b) 0.037 Ω (c) 0.52 Ω (d) 0.41 Ω

46 A sensitive galvanometer, whose coil has a resistance of 50 Ω, has been converted into an ammeter where the opposite graph represents the relation between the intensity of the current that passes through the shunt resistance (I_s) when the galvanometer shows full-scale deflection and the reciprocal of the shunt resistance $(\frac{1}{R_s})$, so the maximum current intensity that can pass through the galvanometer (I_g) equals ..... .



(a) 2×10^{-3} A (b) 3×10^{-2} A (c) 6×10^{-3} A (d) 9×10^{-2} A

47 An ammeter consists of a galvanometer of resistance R_g and a variable resistance R_s acts as a shunt resistance, the opposite graph represents the relation between the current intensity (I) measured by ammeter and the reciprocal of the shunt resistance $(\frac{1}{R_s})$, then the resistance of the galvanometer is determined from the relation; .....



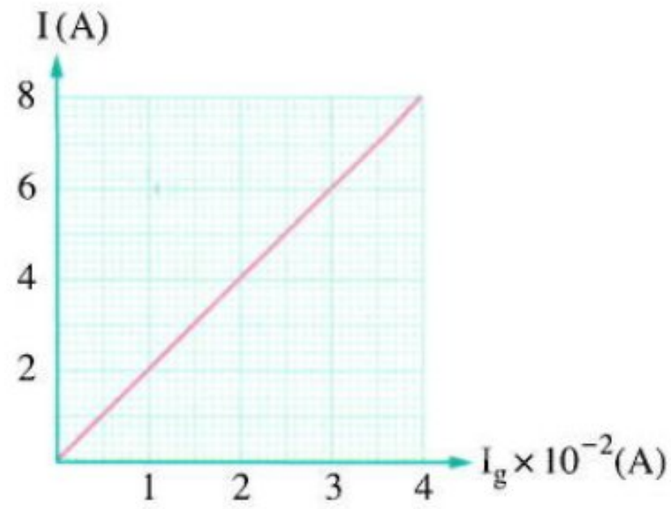
(a) $R_g = \frac{L}{M}$ (b) $R_g = \frac{L}{KM}$

(c) $R_g = \frac{M}{L}$ (d) $R_g = \frac{L+K}{M}$

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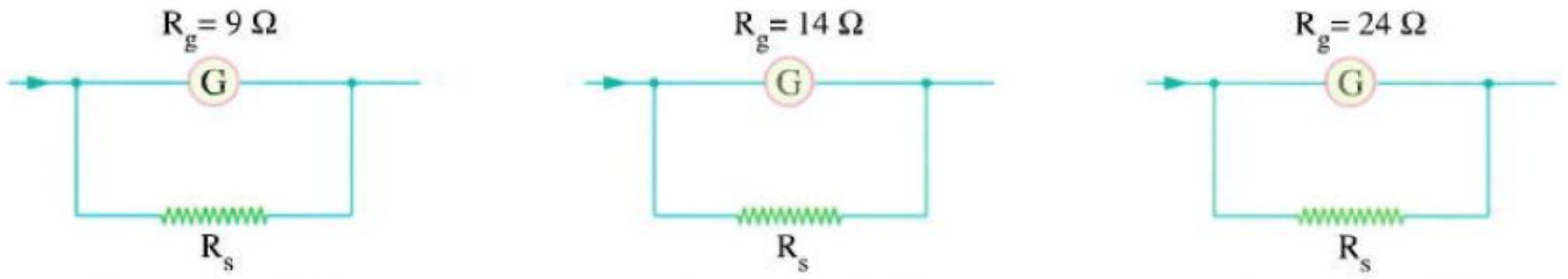
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48 A sensitive galvanometer of resistance 6 Ω is connected to a shunt resistor R_s to be converted into an ammeter and the ammeter is connected in an electric circuit, where the opposite graph represents the relation between the reading of the ammeter (I) and the electric current intensity (I_g) that passes through the coil of the galvanometer, so the value of the shunt resistor R_s is equal to ..... .

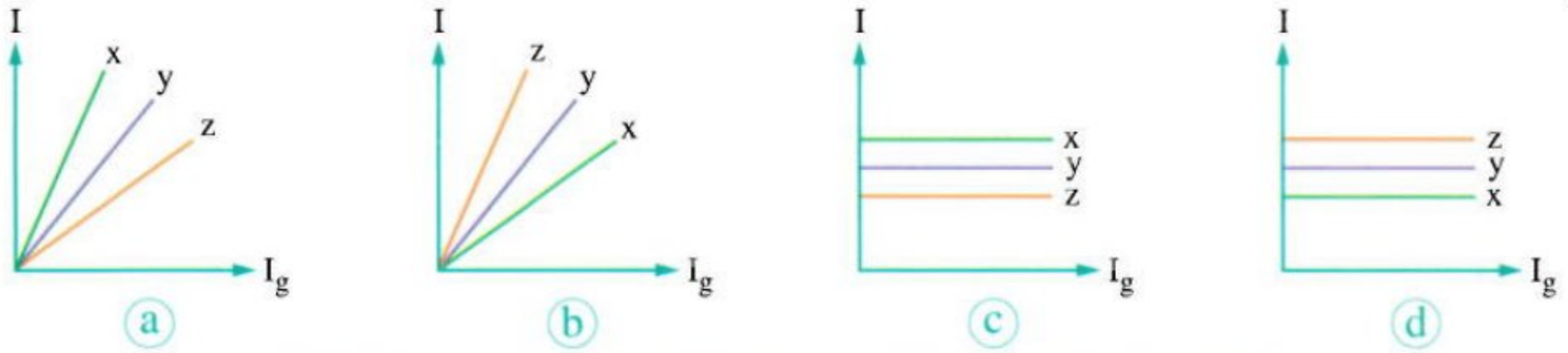


(a) 0.1 Ω (b) 0.03 Ω (c) 0.5 Ω (d) 0.8 Ω

49 Three galvanometers of different resistances, each of them is connected to a shunt resistance R_s one at a time to form three ammeters x, y and z as shown in the following figures:



Which of the following graphs represents the relation between the electric current (I) measured by the ammeter and the electric current that passes through the galvanometer (I_g)?



50 A moving coil galvanometer has a coil of resistance 100 Ω where each one division of its scale is representing an electric current of intensity 25 mA. If its coil is connected to a shunt resistor of 0.05 Ω, so the electric current intensity denoted by one division equals ..... .

(a) 30.06 A (b) 40.01 A (c) 25.02 A (d) 50.025 A

51 * The maximum angle of deflection for a moving coil galvanometer from zero position is 80° and when an electric current of 30 mA passes through the galvanometer, its angle of deflection from the zero position becomes 60°, so:

(i) The sensitivity of the galvanometer equals ..... .

(a) 5 deg/mA (b) 3 deg/mA (c) 2 deg/mA (d) 1 deg/mA

(ii) The maximum current that can be withstood by the galvanometer's coil equals ..... .

(a) 0.02 A (b) 0.04 A (c) 0.06 A (d) 0.08 A

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(iii) The maximum current that can be measured by the device if its coil is connected to a shunt resistance of a value that equals 0.01 of the resistance of its coil equals ..... .

(a) 1.51 A (b) 2.12 A (c) 3.75 A (d) 4.04 A

52 ✳ A galvanometer coil of resistance 20 Ω was connected in parallel to a shunt resistance made of a wire of length 20 cm and resistance 5 Ω, so the galvanometer becomes able to measure a current of maximum value 1 A. Then another wire is used with the same galvanometer as a shunt resistance where this wire was made from the same material and has the same cross-sectional area but has half the length of the first wire, so the maximum current that can be measured by the device becomes ..... .

(a) 1.2 A (b) 1.4 A (c) 1.6 A (d) 1.8 A

53 ✳ A sensitive galvanometer measures an electric current up to a maximum value of 8×10^{-3} A when it is connected to a shunt resistance of 1 Ω and it measures an electric current up to a maximum value of 71×10^{-3} A when it is connected to a shunt resistance of 0.1 Ω, so:

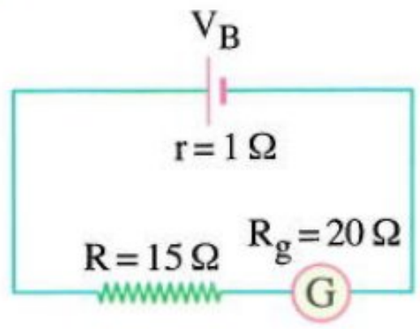
(i) The resistance of the galvanometer's coil equals ..... .

(a) 7 Ω (b) 9 Ω (c) 11 Ω (d) 15 Ω

(ii) The maximum current that can be withstood by the galvanometer's coil equals ..... .

(a) 1×10^{-3} A (b) 2×10^{-3} A (c) 3×10^{-3} A (d) 4×10^{-3} A

54 ✳ The opposite electric circuit consists of battery V_B whose internal resistance is 1 Ω, a constant resistor of 15 Ω and a galvanometer of resistance 20 Ω, hence the ratio between the two intensities of the passing currents in the circuit before and after connecting the galvanometer coil with a shunt resistor of 5 Ω equals ..... .



(a) $\frac{5}{9}$ (b) $\frac{9}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

55 ✳ A galvanometer that has a resistance of 10 Ω where it can be used to measure a current up to a maximum intensity of 40 mA, is connected to a shunt resistance (R_s) then in an electric circuit that contains a resistor of 8 Ω and an electric cell of emf that equals 1.5 V which has a negligible internal resistance. When the circuit is closed, the pointer of the galvanometer deflects to $\frac{3}{4}$ of its graduation scale, so the value of the shunt resistance equals ..... .

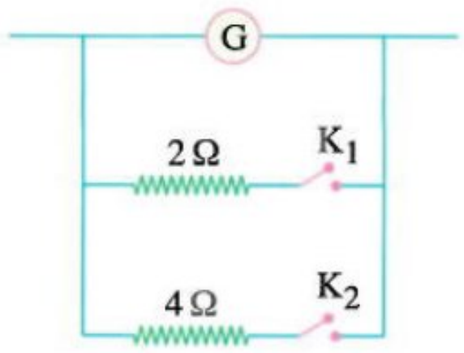
(a) 7.6 Ω (b) 5 Ω (c) 2.5 Ω (d) 0.3 Ω

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56 ✱ In the opposite figure, a galvanometer that can be connected to one of shunt resistances or both, when switch K_1 only is closed, the sensitivity of the device decreases to its quarter and it becomes suitable for measuring a current up to an intensity of 0.5 A, so:



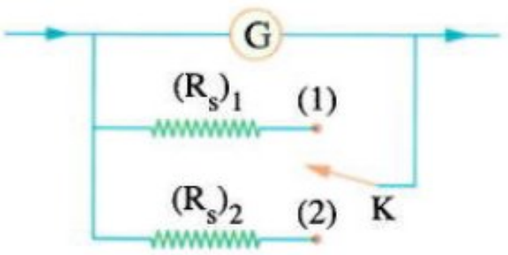
(i) The maximum current that can be measured when closing switch K_2 only equals ..... .

(a) 0.45 A (b) 0.31 A (c) 0.25 A (d) 0.17 A

(ii) The maximum current that can be measured when closing both switches K_1 and K_2 together equals ..... .

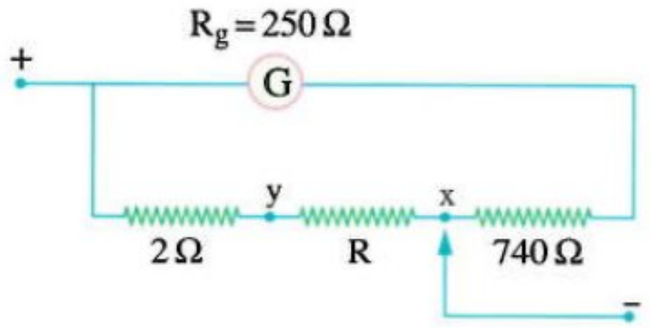
(a) 0.38 A (b) 0.43 A (c) 0.52 A (d) 0.69 A

57 A galvanometer is connected to switch K by two methods as shown in the opposite figure, when the switch is at position (1), 90% of the total current passes through the shunt $(R_s)_1$ and when the switch is at position (2), 99% of the total current passes through the shunt $(R_s)_2$, then the ratio between the two shunt resistances $\left(\frac{(R_s)_1}{(R_s)_2}\right)$ equals ..... .



(a) 10 (b) 11 (c) 9 (d) 99

58 A galvanometer coil of resistance 250 Ω and the maximum current which can be measured is 0.1 A and it has been connected to multiple resistors to form an ammeter, as shown in the opposite figure. When the slider is positioned at point "x" the maximum reading of the device is 10 A. So, the maximum reading of the device when the slider is moved to point "y" is ..... .



(a) 20 A (b) 30 A (c) 40 A (d) 50 A

59 ✱ An electric circuit consists of a battery of negligible internal resistance, a constant resistor of 350 Ω and a galvanometer that is connected in parallel to a shunt resistance of 20 Ω. If the shunt resistance is replaced with another of 30 Ω resistance and to keep the galvanometer deflection unchanged it is required to replace the constant resistor with another of 450 Ω resistance, hence the resistance of the galvanometer equals ..... .

(a) 20 Ω (b) 40 Ω (c) 60 Ω (d) 80 Ω

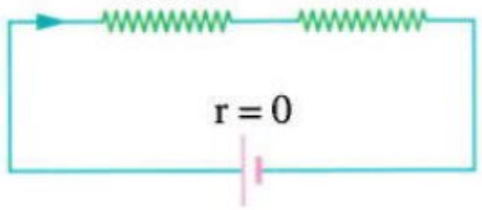
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The voltmeter

60 Three voltmeters (A, B, C) have the same measuring range while the resistances of them respectively are 1000 Ω, 4000 Ω and 8000 Ω, so the voltmeter which will have the highest accuracy of measuring the potential difference between the same two points in the same circuit is ..... .

(a) the voltmeter A
(b) the voltmeter B
(c) the voltmeter C
(d) all of them have the same accuracy

61 In the opposite electric circuit, the voltage across the 300 Ω resistor is V_1. When a voltmeter of resistance 3000 Ω is connected across the 300 Ω resistor terminals, its reading is V_2. When a voltmeter of 6000 Ω resistance is connected across the same 300 Ω resistor, its reading is V_3. Which of the following relation is correct in terms of V_1, V_2, V_3?



(a) $V_1 = V_2 = V_3$
(b) $V_1 > V_2 > V_3$
(c) $V_1 > V_3 > V_2$
(d) $V_1 < V_2 < V_3$

62 ✳ A galvanometer shows full-scale deflection when a current of 50 μA passes through it. If it has been converted into a voltmeter that can read up to 10 V, so:

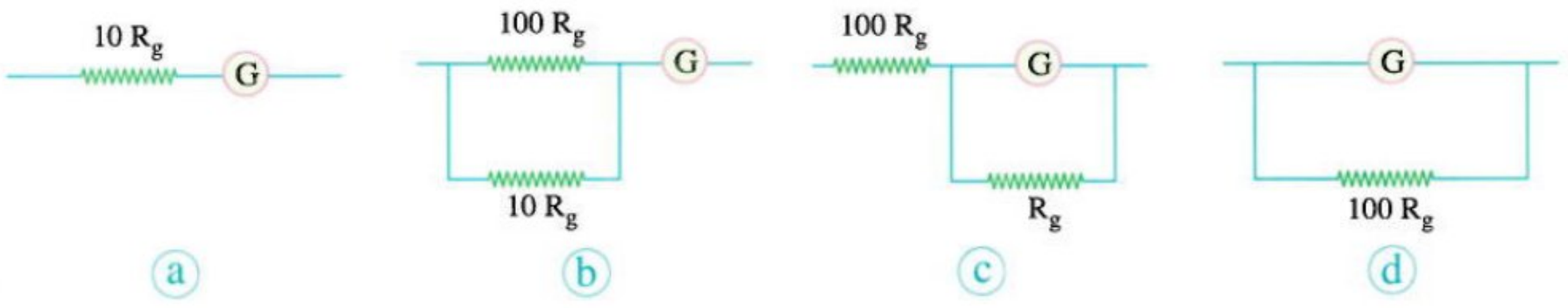
(i) The total resistance of the voltmeter equals ..... .

(a) 100×10^3 Ω (b) 200×10^3 Ω (c) 300×10^3 Ω (d) 400×10^3 Ω

(ii) The value of the multiplier resistance if the resistance of the galvanometer coil is 1 kΩ equals ..... .

(a) 111×10^3 Ω (b) 150×10^3 Ω (c) 199×10^3 Ω (d) 245×10^3 Ω

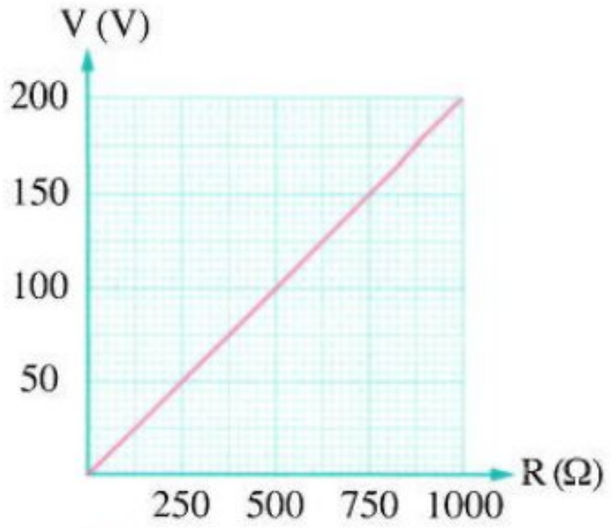
63 Four similar galvanometers each of resistance R_g and also each of them is connected to a resistor or more than one resistor as in the following figures, which of the following voltmeters can withstand higher potential difference?



(a) (b) (c) (d)

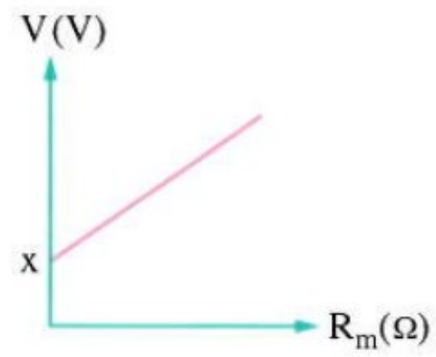
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64 A sensitive galvanometer that can measure a current of maximum limit I_g, was connected with different multiplier resistors each one at a time to convert it into a voltmeter and the opposite graph represents the relation between the maximum potential difference (V) that can be measured by the voltmeter and the total resistance of the voltmeter (R), so the value of I_g is ..... .



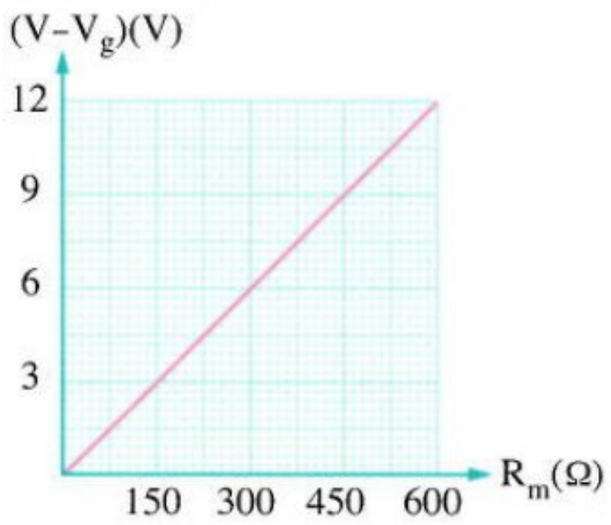
(a) 0.1 A (b) 0.2 A (c) 0.25 A (d) 0.5 A

65 The opposite graph represents the relation between the total potential difference (V) between the terminals of a voltmeter and the value of the multiplier resistance (R_m) in the voltmeter, so the quotient of $\frac{\text{value of x}}{\text{slope}}$ represents ..... .



(a) R_g (b) V_m (c) $\frac{1}{I_m}$ (d) V_g

66 The opposite graph represents the variation of the difference of the maximum voltages that can be measured by the galvanometer before and after connecting the voltage multipliers ($V - V_g$) with the change of the voltage multiplier (R_m);



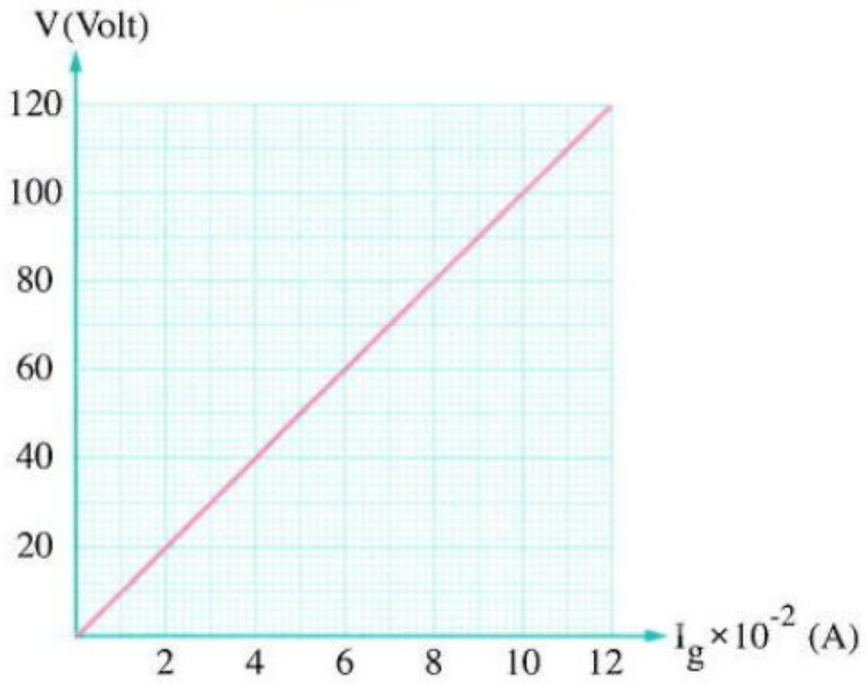
(i) So the maximum current intensity that can be withstood by the galvanometer before connecting the voltage multiplier equals ..... .

(a) 0.01 A (b) 0.02 A (c) 0.03 A (d) 0.04 A

(ii) If the maximum potential difference that can be withstood by the galvanometer's coil before connecting the voltage multiplier is 1 V, so the galvanometer's coil resistance equals ..... .

(a) 30 Ω (b) 50 Ω (c) 80 Ω (d) 100 Ω

67 * A sensitive galvanometer contains a coil whose resistance is 50 Ω where the maximum current that it can withstand is 0.12 A. It is connected to a multiplier resistance (R_m) to be converted into a voltmeter and the opposite graph represents the relation between the voltmeter reading (V) and the intensity of the current which passes through its coil (I_g), so:



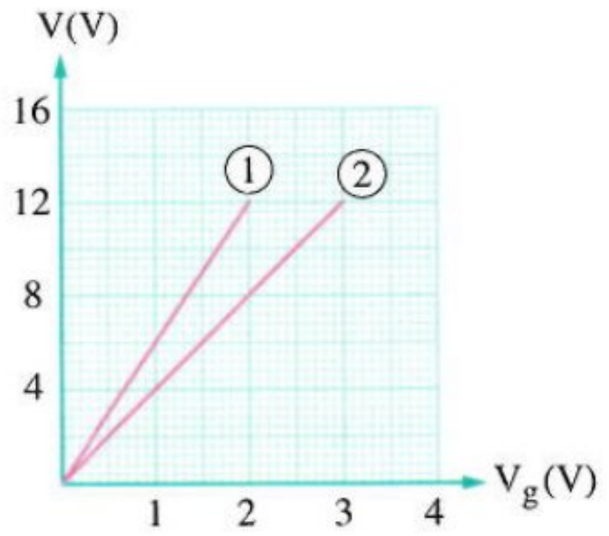
(i) The resistance of the voltage multiplier (R_m) equals ..... .

(a) 125 Ω (b) 240 Ω
(c) 460 Ω (d) 950 Ω

(ii) The maximum voltage that can be measured by the voltmeter equals ..... .

(a) 90 V (b) 120 V (c) 210 V (d) 180 V

68 The maximum voltage that can be measured by a sensitive galvanometer is V_g and it is connected separately to two voltage multipliers 1 kΩ and R to convert it into two voltmeters (1) and (2), respectively. The opposite graph represents the relation between the voltage difference across each voltmeter (V) and the voltage difference across the galvanometer (V_g). Then the value of the voltage multiplier R is equal to ..... .



(a) 200 Ω (b) 400 Ω (c) 600 Ω (d) 800 Ω

69 If the multiplier resistance in a voltmeter is ten times as large as the galvanometer's resistance and the maximum voltage that can be withstood by the galvanometer is V_g, then the maximum voltage that can be measured by the voltmeter equals ..... .

(a) 10 V_g (b) 9 V_g (c) 0.1 V_g (d) 11 V_g

70 A voltmeter consists of a galvanometer of resistance R and a multiplier resistance of 50 R, so which of the following ratios equals 0.02?

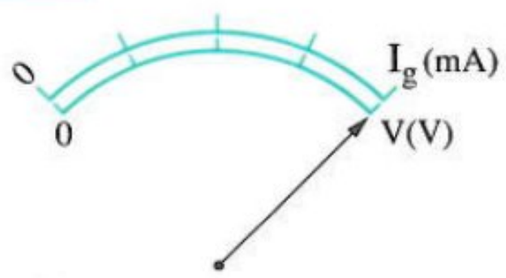
(a) The ratio between the current intensity that passes in the galvanometer and that in the voltmeter.

(b) The ratio between the current intensity that passes in the galvanometer and that in multiplier resistance.

(c) The ratio between the voltage across the multiplier resistance and that across the voltmeter.

(d) The ratio between the voltage across the galvanometer and that across the multiplier resistance.

71 The opposite figure shows the graduation scale of a galvanometer after its calibration into a voltmeter scale, what does the ratio between the reading of the voltmeter and that of the galvanometer $\left(\frac{V}{I_g}\right)$ represent?



(a) $R_m + R_g$ (b) $R_m - R_g$ (c) $R_m R_g$ (d) $\frac{R_m}{R_g}$

72 A galvanometer of resistance 100 Ω that can withstand a maximum current of 0.01 A is required to be converted into a voltmeter, then:

(i) The multiplier resistance that makes it measure a potential difference up to 5 V is ..... .

(a) 5 Ω (b) 100 Ω (c) 400 Ω (d) 500 Ω

(ii) The value of the maximum potential difference that can be measured by the device when it is connected to a multiplier resistance of 900 Ω is ..... .

(a) 0.9 V (b) 9 V (c) 10 V (d) 90 V

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73 ✱ A galvanometer shows full-scale deflection when an electric current of intensity 0.02 A passes through it when the potential difference between its two terminals is 5 V, so:

(i) The value of the multiplier resistance that makes the device suitable to measure a potential difference of 150 V equals ..... .

(a) 2916 Ω (b) 4374 Ω (c) 5841 Ω (d) 7250 Ω

(ii) The resistance of the galvanometer coil equals ..... .

(a) 110 Ω (b) 250 Ω (c) 315 Ω (d) 520 Ω

74 In a voltmeter device, the ratio between the electric current intensity which passes in the galvanometer coil and that which passes in the multiplier resistance that is connected to it is always ..... one.

(a) greater than (b) equal to (c) less than (d) indeterminable

75 As the value of the multiplier resistance of a voltmeter increases, ...

(a) the total resistance of the device decreases

(b) the sensitivity of the device increases

(c) the range of the device to measure the potential difference decreases

(d) the accuracy of the device in measuring the potential difference increases

76 A voltmeter of resistance 2000 Ω can measure a maximum potential difference of 2 V. If it is connected to a multiplier resistor R_m, its measuring range increases by 8 V, then the value of R_m is ..... .

(a) 2000 Ω (b) 4000 Ω (c) 6000 Ω (d) 8000 Ω

77 A voltmeter has a resistance of 500 Ω, each division of its graduation scale denotes 0.1 V, so the value of the multiplier resistance that makes each division of its scale denotes 1 V equals ..... .

(a) 5000 Ω (b) 2400 Ω (c) 4500 Ω (d) 2700 Ω

78 ✱ A galvanometer of resistance 5 Ω measures electric currents up to a maximum value of 20 mA, so:

(i) The maximum current that can be measured by the device if it is connected with a shunt resistance of 0.1 Ω equals ..... .

(a) 0.96 A (b) 1.02 A (c) 3.16 A (d) 4.05 A

(ii) The multiplier resistance which could be connected to the galvanometer, so it works as a voltmeter that measures a potential difference up to a maximum value of 5 V equals ..... .

(a) 100 Ω (b) 175 Ω (c) 245 Ω (d) 332 Ω

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79 The resistance of the coil of a moving coil galvanometer is 18 Ω, so:

(i) The value of the shunt resistance that allows $\frac{1}{3}$ of the total current to pass through the galvanometer coil equals ..... .

(a) 3 Ω (b) 6 Ω (c) 9 Ω (d) 12 Ω

(ii) The value of the multiplier resistance that makes the galvanometer suitable for measuring a potential difference up to 10 times as large as the potential difference across its coil equals ..... .

(a) 162 Ω (b) 81 Ω (c) 324 Ω (d) 202.5 Ω

80 ✳ When a moving coil galvanometer is connected to a shunt resistance of 0.5 Ω, it becomes suitable for measuring currents up to a maximum intensity of 0.11 A and when it is connected to a voltage multiplier of resistance 245 Ω, it becomes suitable for measuring a potential difference up to a maximum voltage of 2.5 V, so:

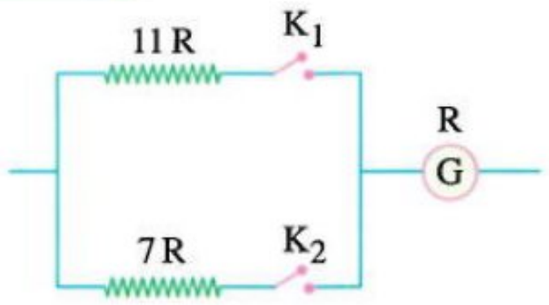
(i) The maximum current (I_g) that can be withstood by the galvanometer coil equals ..... .

(a) 0.5 A (b) 0.25 A (c) 0.01 A (d) 0.005 A

(ii) The galvanometer resistance equals ..... .

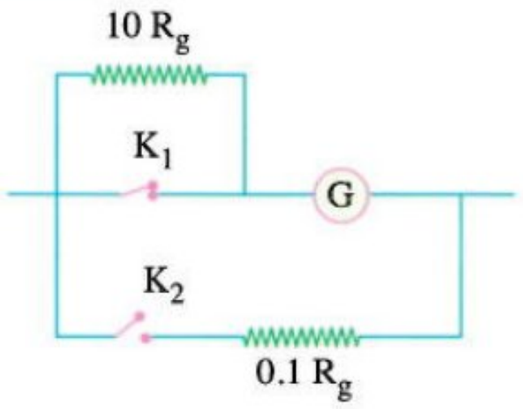
(a) 5 Ω (b) 3 Ω (c) 2.5 Ω (d) 1.5 Ω

81 In the opposite figure, a galvanometer of resistance R is connected to voltage multipliers 7 R and 11 R. When switch K_1 is closed only, the maximum reading of the device for measuring voltage is 24 V. When switch K_2 is closed only, the maximum reading of the device for measuring voltage decreases by an amount of ..... .



(a) 6 V (b) 8 V (c) 12 V (d) 16 V

82 ✳ In the opposite figure, a part of a circuit that contains a galvanometer of resistance R_g where the maximum current it can withstand is I_g. When the switch K_1 is opened, it becomes a voltmeter that can measure a potential difference of maximum value 11 V and when both switches K_1, K_2 are closed together, the galvanometer gets converted into an ammeter that can measure a current of maximum intensity 0.11 A, so the values of I_g, R_g are ..... .

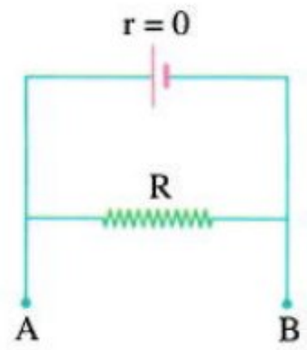


	I_g	R_g
(a)	0.1 A	10 Ω
(b)	0.1 A	100 Ω
(c)	0.01 A	10 Ω
(d)	0.01 A	100 Ω

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83 Two voltmeters X and Y contain identical galvanometers and different multiplier resistances, so what is the correct statement that describes the motions of their pointers when each of them is connected individually between the two points A and B in the opposite circuit?

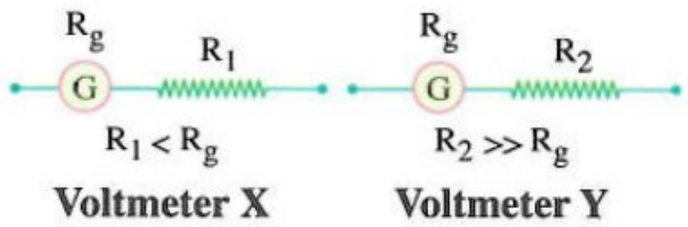


(a) The pointer of device X deflects with a larger angle.

(b) The pointer of device Y deflects with a larger angle.

(c) Both devices' pointers deflect with the same angle.

(d) Both devices' pointers don't deflect.



84 ✱ An electric circuit contains a 10 Ω resistor which is connected in parallel with a voltmeter of 50 Ω resistance. When an electric current of intensity 0.6 A passes in the circuit, the voltmeter shows full scale deflection, so:

(i) The voltmeter reading is ..... .

(a) 2 V (b) 3 V (c) 5 V (d) 9 V

(ii) The maximum potential difference that can be measured by the voltmeter if its coil is connected in series with a resistance of 4950 Ω equals ..... .

(a) 100 V (b) 250 V (c) 370 V (d) 500 V

85 ✱ An electric circuit contains a constant resistor of 6 Ω across which a voltmeter of 30 Ω resistance is connected in parallel. When a current of 0.2 A passes in the circuit, the voltmeter shows full-scale deflection, so if a resistance of 144 Ω is connected in series with the voltmeter and a current of the same intensity passes in the circuit:

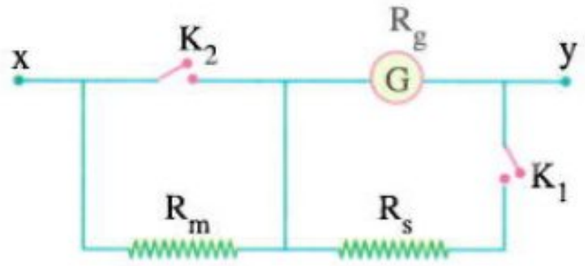
(i) The reading of the voltmeter in this case equals ..... .

(a) 3.48 V (b) 2.34 V (c) 1.16 V (d) 0.98 V

(ii) The maximum potential difference that can be measured by the device after connecting it to the multiplier resistance equals ..... .

(a) 7.2 V (b) 9.6 V (c) 11.2 V (d) 5.8 V

86 An electric device xy consists of a galvanometer, multiplier resistance (R_m), shunt resistance (R_s) and two switches K_1, K_2 as shown in the opposite figure, when the two switches are opened, the sensitivity of the device to measure the potential difference decreases to quarter but when the two switches are closed, the sensitivity of the device to measure the current decreases to quarter, then the ratio of $\left(\frac{R_m}{R_s}\right)$ equals ..... .



(a) 3 (b) 4 (c) 9 (d) 16

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87 ✳ A sensitive galvanometer, the resistance of its coil is 4 Ω, measures a current up to a maximum value of 1 mA, its coil has been connected in parallel to a resistance of 1 Ω where they form one device and that device is connected in series to a resistor of 999.2 Ω to form a voltmeter, so the maximum potential difference that can be measured by this voltmeter equals ..... .

(a) 2 V (b) 5 V (c) 7 V (d) 9 V

88 A milli-ammeter gives full scale deflection when a current of 10 mA passes through its coil. If the device consists of a resistance of 0.2 Ω that is connected in parallel to a galvanometer of resistance 33 Ω, then the value of the resistance which is required to be connected in series to convert the milli-ammeter into a voltmeter that measures a potential difference up to 10 V equals ..... .

(a) 880.2 Ω (b) 950.3 Ω (c) 999.8 Ω (d) 1250.4 Ω

89 ✳ An electric circuit contains an electric cell of emf 10 V whose internal resistance is negligible, connected with two resistors of 16 Ω, 40 Ω in series and when a voltmeter is connected in parallel with the 40 Ω resistor, its pointer has deflected to 6 V:

(i) So, the resistance of the voltmeter equals ..... .

(a) 25 Ω (b) 53 Ω (c) 60 Ω (d) 67 Ω

(ii) If the maximum reading of the voltmeter is 7.5 V, the value of the shunt resistance that leads to convert it into an ammeter measuring a current up to a maximum value of 5 A equals ..... .

(a) 0.6 Ω (b) 1.54 Ω (c) 2.17 Ω (d) 3.72 Ω

90 A galvanometer of resistance R_g is connected separately to voltage multipliers 950 Ω and 3950 Ω to convert it into two voltmeters (1) and (2). If the ratio between the maximum voltage that each voltmeter can measure $\left(\frac{V_1}{V_2}\right)$ is $\frac{1}{4}$, then the resistance of the galvanometer R_g is equal to ..... .

(a) 20 Ω (b) 36 Ω (c) 50 Ω (d) 100 Ω

91 A galvanometer has a maximum measurable voltage V_g. When it is connected separately to voltage multipliers R_1 and R_2, the maximum voltage each device can measure is nV_g and mV_g, respectively. The ratio of $\left(\frac{R_1}{R_2}\right)$ is equal to ..... .

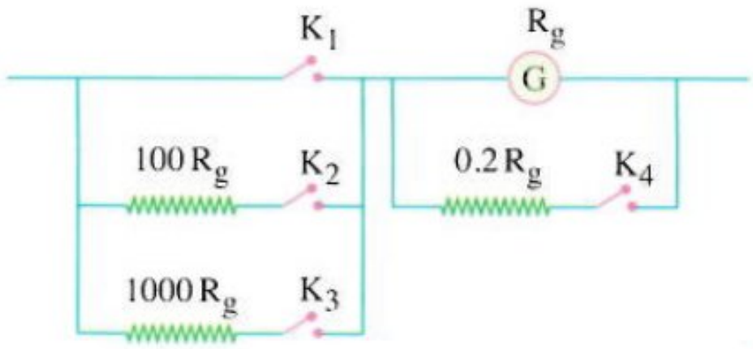
(a) $\frac{n-1}{m-1}$ (b) $\frac{n+1}{m+1}$ (c) $\frac{m-1}{n-1}$ (d) $\frac{m+1}{n+1}$

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92 In the opposite figure, to convert the galvanometer into a voltmeter to measure the maximum potential difference, so the two switches must be closed are ..... .



(a) K_1, K_4 (b) K_1, K_3
(c) K_3, K_4 (d) K_2, K_4

The ohmmeter

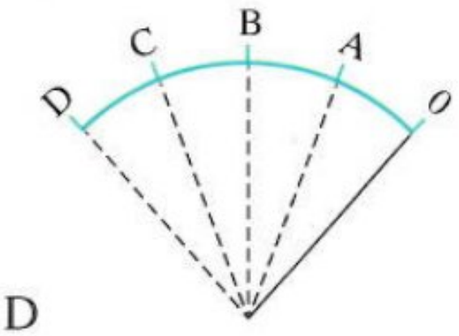
93 The ohmmeter pointer deflects to its maximum scale when the ohmmeter circuit is closed, so the measured resistance is ..... .

(a) very large (b) small (c) zero (d) infinite

94 A resistor R is connected between the terminals of an ohmmeter, hence its pointer has deflected to half its current scale, so the value of the resistor R is ..... .

(a) infinite
(b) equal to the ohmmeter resistance
(c) equal to zero
(d) greater than the ohmmeter resistance

95 The opposite figure represents equal divisions of the ohmmeter scale whose pointer is adjusted at zero position.



(i) Which of the shown marks on the scale can represent the resistance of ohmmeter?

(a) A (b) B (c) C (d) D

(ii) When the pointer settled at the mark C, it means that the value (C) is ..... .

(a) less than the resistance of ohmmeter
(b) equal to the summation of the ohmmeter's resistance and the resistance which is connected to its terminals
(c) double the resistance of ohmmeter
(d) equal to the resistance that is connected to the terminals of ohmmeter

96 * A milli-ammeter of 4 Ω resistance has a coil that can withstand a current up to a maximum intensity of 16 mA, it is wanted to be converted into an ohmmeter using a dry cell of emf 1.5 V whose internal resistance is 1.75 Ω, so:

(i) The value of the standard resistance which is required to be used equals ..... .

(a) 25 Ω (b) 43 Ω (c) 88 Ω (d) 95 Ω

(ii) The external resistance that makes its pointer deflect to 10 mA equals ..... .

(a) 311.75 Ω (b) 150.9 Ω (c) 112.5 Ω (d) 56.25 Ω

(iii) The intensity of the electric current that passes through it if it is connected to an external resistance of 300 Ω equals ..... .

(a) 2×10^{-3} A (b) 3.8×10^{-3} A (c) 5×10^{-3} A (d) 6.4×10^{-3} A

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97 ✳ A galvanometer of coil resistance 250 Ω that shows a full-scale deflection when a 400 μA current passes through its coil is connected to an electric cell of emf 1.5 V and of negligible internal resistance, a constant resistance of 3000 Ω and a variable resistance (R_V), so:

(i) The obtained value from the variable resistor, so that the galvanometer would be converted into an ohmmeter, equals ..... .

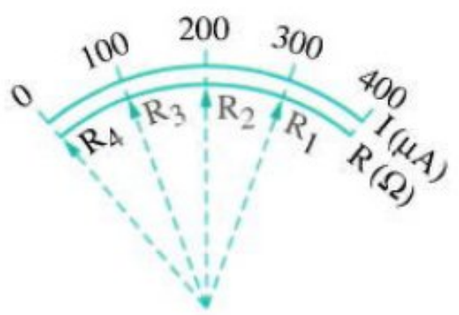
(a) 2000 Ω (b) 1500 Ω (c) 1000 Ω (d) 500 Ω

(ii) The value of the resistance, that if connected to the terminals of the ohmmeter makes the pointer deflect to the quarter of its scale, equals ..... .

(a) 3750 Ω (b) 7500 Ω (c) 15315 Ω (d) 11250 Ω

---

98 In the opposite figure, the ohms graduation scale is added to the ammeter graduation scale, so if the total internal resistance of the ohmmeter is 3750 Ω and the maximum value of current intensity on its scale is 400 μA:



(i) The resistance of R_1 equals ..... .

(a) 5317 Ω (b) 3750 Ω (c) 1250 Ω (d) 999 Ω

(ii) The resistance of R_2 equals ..... .

(a) 1250 Ω (b) 2500 Ω (c) 3750 Ω (d) 5412 Ω

(iii) The resistance of R_3 equals ..... .

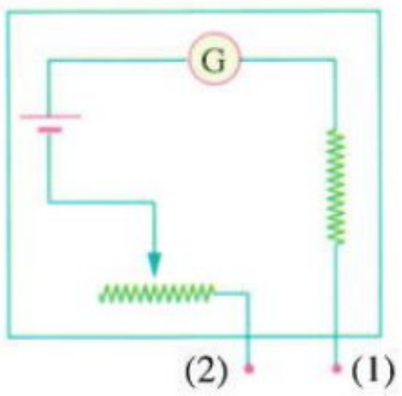
(a) 3750 Ω (b) 7500 Ω (c) 9713 Ω (d) 11250 Ω

(iv) The expected resistance of R_4 and the reason for this expectation are ..... .

	The value	The reason
(a)	∞	because the current intensity vanishes
(b)	∞	because the current intensity becomes maximum
(c)	0	because the current intensity vanishes
(d)	0	because the current intensity becomes maximum

---

99 The opposite figure shows the components of an ohmmeter where the full scale deflection of the galvanometer pointer is 600 μA at connecting the ends of the ohmmeter (1), (2) together, if a resistance R_x which equals double of the total resistance of the circuit was inserted between the terminals of the ohmmeter, then, the reading of the galvanometer pointer equals ..... .



(a) 200 μA (b) 300 μA (c) 600 μA (d) 1200 μA

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100 An ohmmeter of total resistance R_o contains a battery of emf V_B that has a negligible internal resistance, when an unknown resistance R has been connected between the terminals of the ohmmeter, the pointer of the device has deflected to $\frac{1}{5}$ of its current scale, then the value of the ohmmeter's resistance (R_o) equals ..... .

(a) 5 R (b) $\frac{R}{5}$ (c) 4 R (d) $\frac{R}{4}$

101 If a resistance of 200 Ω makes the ohmmeter's pointer deflects to $\frac{1}{2}$ the current scale, then the resistance that makes it deflect to $\frac{1}{3}$ of the current scale is ..... .

(a) 300 Ω (b) 400 Ω

(c) 600 Ω (d) 1200 Ω

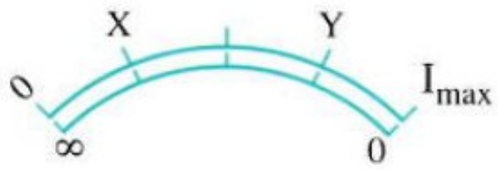
102 The opposite figure represents an equally divided ohmmeter scale. When an external resistance is connected between the two terminals of the device, its pointer has deflected to position x on the scale of the current, so the value of this resistance is ..... the resistance of the ohmmeter.



(a) one third (b) half

(c) double (d) three times

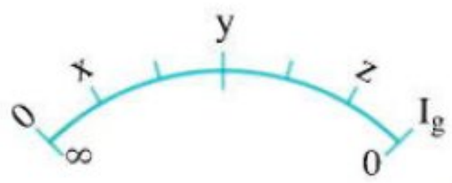
103 ✱ The opposite figure shows an equally divided ohmmeter scale. When this ohmmeter is used to measure an unknown resistance R, the pointer of the device is deflected to the position X on the scale, then the external resistance that makes the pointer of the device deflects to the position Y on the scale is ..... .



(a) $\frac{R}{3}$ (b) 3 R

(c) $\frac{R}{9}$ (d) $\frac{3R}{4}$

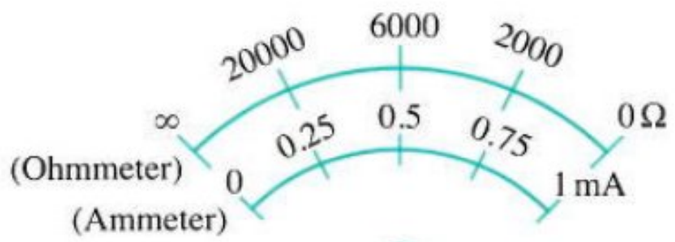
104 The opposite figure represents an equal divisions of a calibrated ohmmeter where the position y represents exactly the middle of the scale, then the ratio between the values of the resistance that is connected to the terminals of the ohmmeter $\left(\frac{R_x}{R_z}\right)$ when the pointer settles at the two positions x, z is ..... .



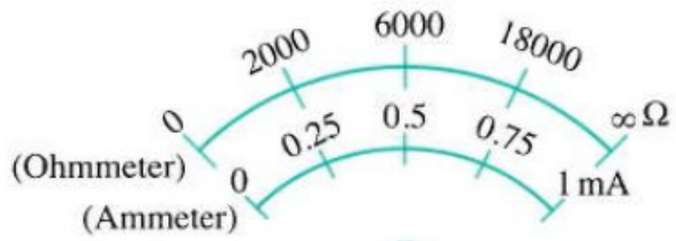
(a) 4 (b) 9 (c) 16 (d) 25

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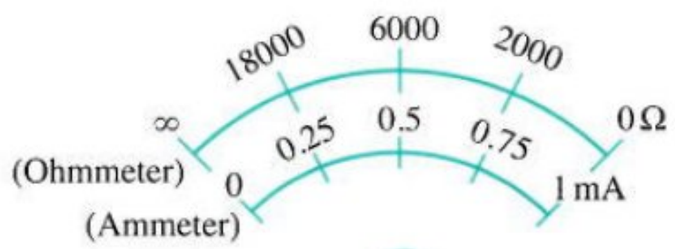
105 An ohmmeter consists of an ammeter, a standard fixed resistance and a 6 V battery. It shows a full-scale deflection when a current of 1 mA flows through it, if an ohm graduation scale is added to the ammeter scale, the correct figure that represents the scales of the current and resistance will be ..... .



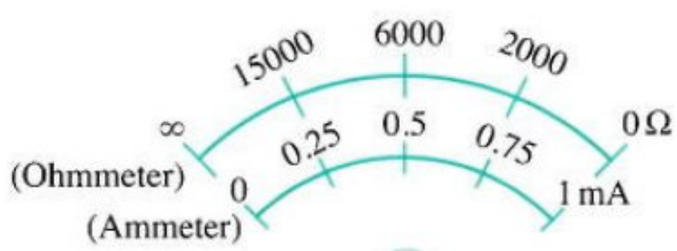
a



b

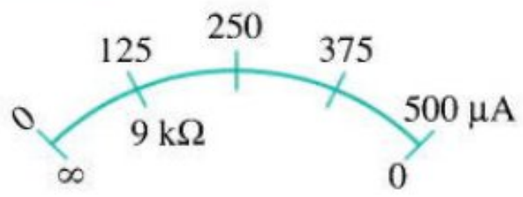


c



d

106 ✳ The opposite figure represents equal divisions on the ohmmeter graduation scale. Using the shown data:



(i) The ohmmeter resistance equals ..... .

(a) 9000 Ω (b) 6000 Ω

(c) 3000 Ω (d) 1000 Ω

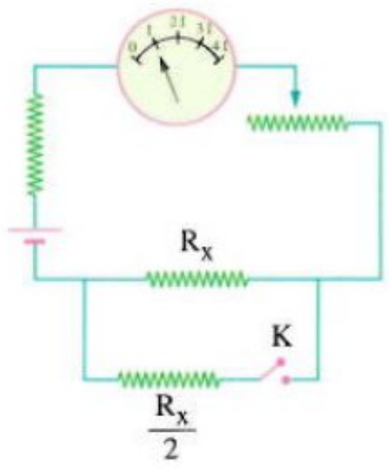
(ii) The electromotive force of the dry cell in the ohmmeter equals ..... .

(a) 1.5 V (b) 2 V (c) 2.5 V (d) 3 V

107 ✳ When 300 Ω resistor is connected to an ohmmeter, its pointer deflects to $\frac{1}{4}$ of its scale, so the resistance that makes its pointer deflect to $\frac{1}{6}$ of its scale equals ..... .

(a) 1500 Ω (b) 1000 Ω (c) 500 Ω (d) 250 Ω

108 In the shown ohmmeter circuit, when the switch K is closed, the pointer becomes ..... .



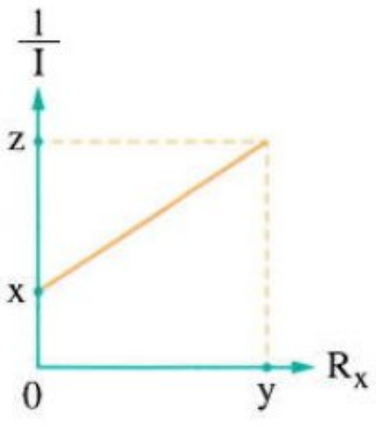
(a) at 2 I

(b) between 2 I, I

(c) at 3 I

(d) between 3 I, 2 I

109 An ohmmeter has been calibrated, and its total resistance is R_0. The opposite graph represents the relation between the reciprocal of the current $\left(\frac{1}{I}\right)$ passing through the ohmmeter circuit and the external resistance R_x connected across its terminals. Thus, the resistance of the ohmmeter R_0 is equal to ..... .



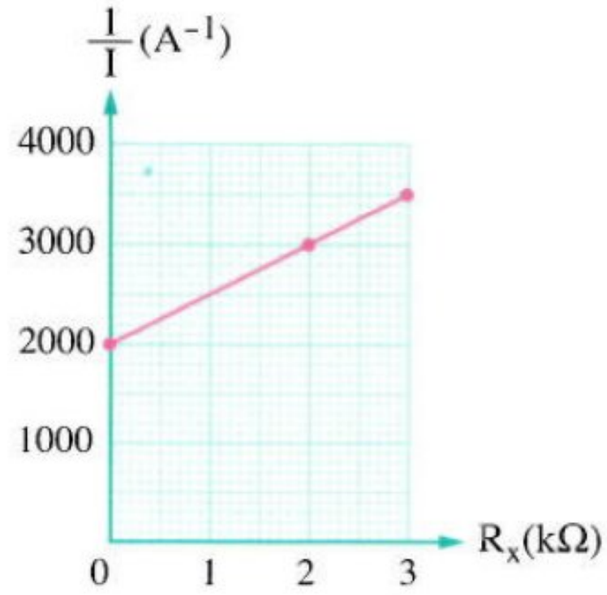
(a) $\frac{zy}{z-x}$ (b) $\frac{xy}{z-x}$ (c) $\frac{z-x}{zy}$ (d) $\frac{z-x}{xy}$

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110 An ohmmeter has been calibrated, and its total resistance is R_0. The opposite graph represents the relation between the reciprocal of the current $\left(\frac{1}{I}\right)$ passing through the ohmmeter circuit and the external resistance (R_x) that is connected across its terminals. Thus,



(i) The electromotive force (emf) of the battery used in the ohmmeter is equal to ..... .

(a) 0.5 V (b) 1 V
(c) 1.5 V (d) 2 V

(ii) The resistance of ohmmeter (R_0) equals ..... .

(a) 2 kΩ (b) 4 kΩ (c) 6 kΩ (d) 8 kΩ

Second Essay questions

The moving coil galvanometer

1 **In the moving coil galvanometer, explain:**

(a) The galvanometer pointer deflects when a current passes through it, then it stops at a certain reading.

(b) When the current is cut, the pointer of the galvanometer returns to the zero position.

2 **Mention two characteristics** to be considered during making a moving coil galvanometer, so its scale becomes uniform.

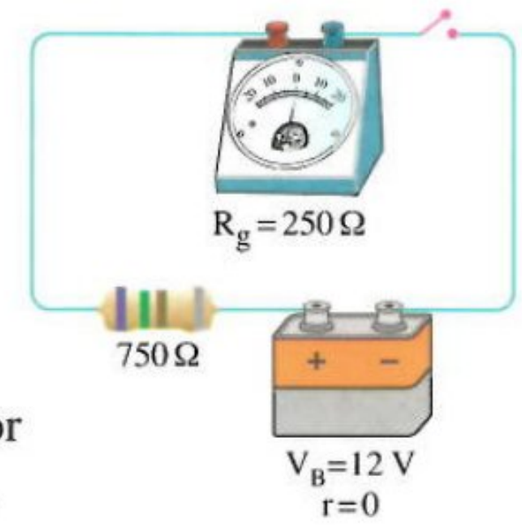
3 **Give reason:** During the rotation of a galvanometer coil that carries a current of intensity I, the value of the magnetic torque acting on it remains unchanged.

4 The scale of a moving coil galvanometer is divided into 20 divisions where the current indication of each division is 200 μA. **Calculate** the intensity of the current that deflects the galvanometer pointer to the quarter of its scale.

5 **Why** is the galvanometer reading for the intensity of the current passing in an electric circuit always less than its intensity when the galvanometer is not part of it?

The ammeter

6 A galvanometer is connected in an electric circuit as in the opposite figure, if the maximum reading on its scale is 20 μA:



(a) **Explain why** such a galvanometer is not suitable for measuring the intensity of the current passing in that circuit when the switch is closed.

(b) **Suggest a method** that makes such galvanometer suitable for measuring the intensity of the current passing in that circuit.

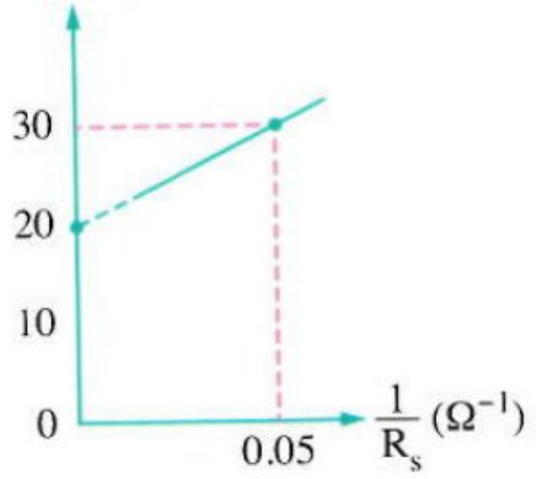
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7 A galvanometer can measure an electric current of maximum intensity 0.01 A. If its coil resistance is 30 Ω, **calculate** the resistance of the shunt which is required to convert it into an ammeter that can measure a current of maximum intensity 1 A and also the total resistance of the ammeter in that case.

8 A galvanometer of resistance R_g, the maximum reading on its scale is I_g is connected to a shunt of resistance R_s, so the total resistance of the device becomes $\frac{R_g}{4}$ and the maximum current that it can measure becomes I. **Determine the ratio between:**

(a) The two resistances $(\frac{R_s}{R_g})$. **(b)** The two currents $(\frac{I}{I_g})$.

9 A sensitive galvanometer whose coil can be connected to different shunt resistances each separately where the opposite graph represents the relation between the maximum current intensity that the ammeter can measure (I) and the reciprocal of the shunt resistance $(\frac{1}{R_s})$. **Calculate:**



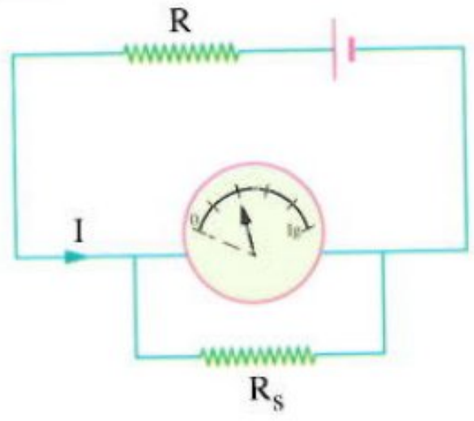
(a) The maximum current intensity that the galvanometer can measure (I_g).

(b) The resistance of the galvanometer (R_g).

(c) The resistance of the shunt that makes the ammeter measure a current of maximum intensity 0.3 A.

10 When a galvanometer is connected to a shunt of resistance R_s, its maximum measuring range becomes 0.06 A and when it is connected to a shunt resistance of 4 R_s, its maximum measuring range becomes 0.03 A, **so what** is the maximum current intensity that can be measured by the galvanometer before connecting any shunt resistance?

11 A moving coil galvanometer of resistance R_g has a scale divided into five divisions each indicates 0.01 A. When the galvanometer is connected to a shunt of resistance 0.2 R_g and gets embedded in an electric circuit, its pointer has deflected as in the opposite figure. **Calculate:**



(a) The current intensity I that passes in the circuit.

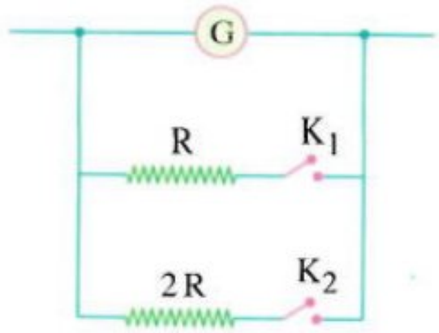
(b) The maximum current intensity that can be measured by the ammeter.

12 An ammeter has a resistance of 24 Ω, **calculate** the resistance of the shunt that is required to decrease the sensitivity of such ammeter to $\frac{1}{9}$ of its value.

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13 In the opposite figure, when switch K_1 is closed, the sensitivity of the device becomes half that of the galvanometer. **Calculate** the ratio between the sensitivity of the device and that of the galvanometer when the switch K_2 only is closed.



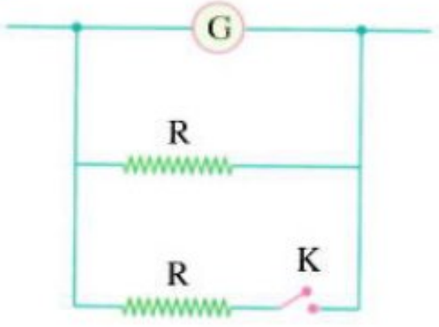
14 **The opposite figure shows the structure of an ammeter. What happens to each of the following when the switch K is closed:**

(a) Resistance of the ammeter?

(b) The maximum current intensity that can be measured by the ammeter?

(c) Accuracy of the ammeter?

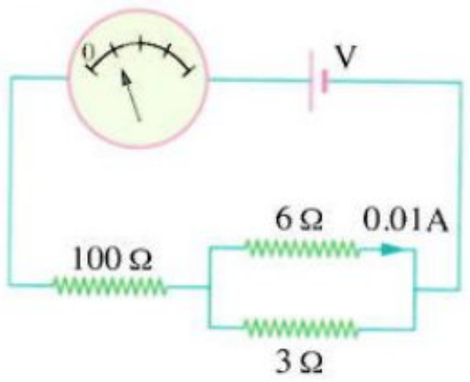
(d) Sensitivity of the ammeter?



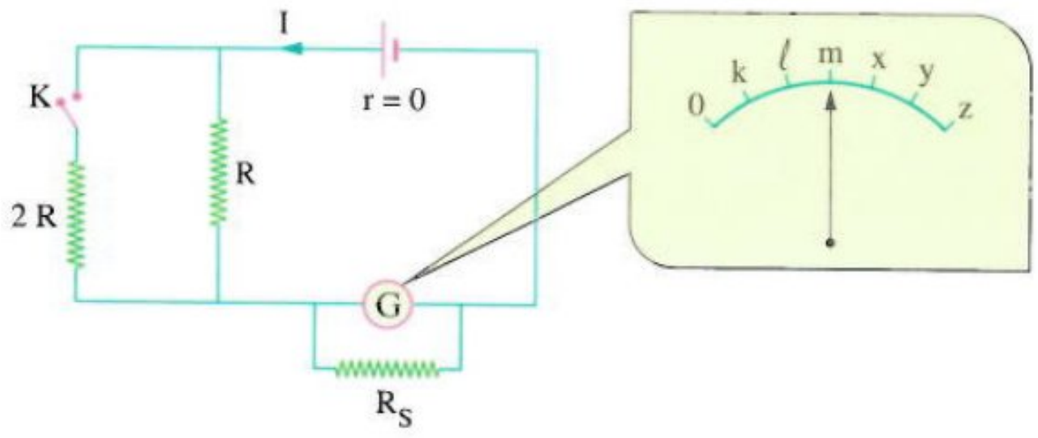
15 In the opposite figure, a galvanometer of resistance 50 Ω is connected to an electric circuit:

(a) **What** is the maximum reading of that galvanometer?

(b) **How** can it be converted into an ammeter that can measure a current of maximum intensity 0.6 A?



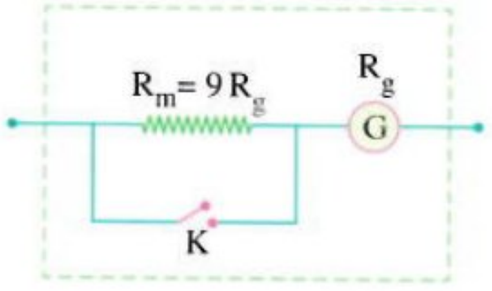
16 The following figure shows the position of the pointer of a galvanometer of resistance R, that is connected to a shunt of resistance 0.5 R and embedded in an electric circuit.



Deduce the deflection position of the pointer when the switch (K) is closed.

The voltmeter

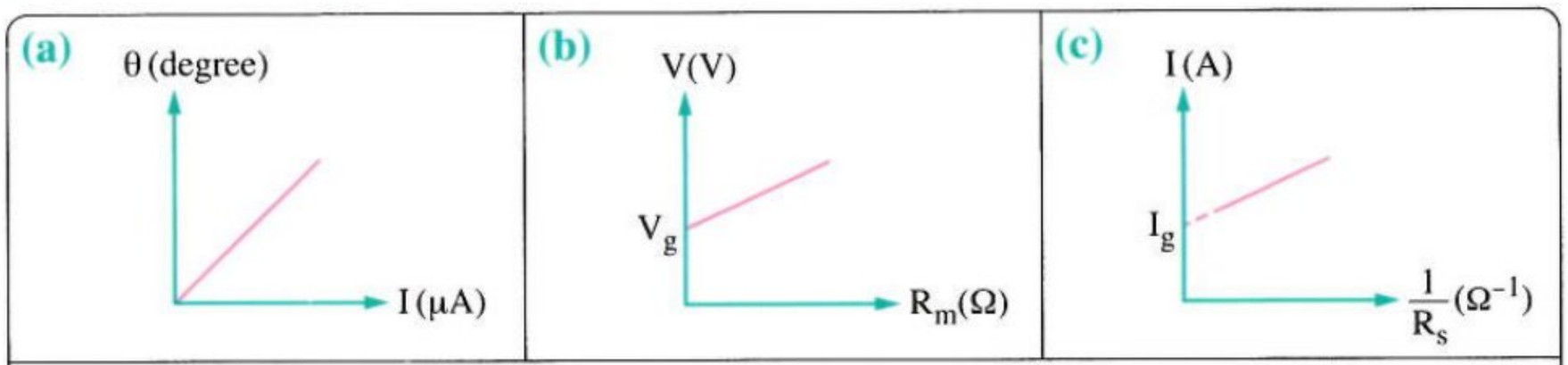
17 The opposite figure represents a device that can measure a potential difference of maximum value 1 V while the switch (K) is opened. **What** is the maximum potential difference that can be measured by the device when the switch (K) is closed?



18 **Explain:** The voltmeter is connected in parallel between the terminals of the conductor across which it is required to measure the potential difference.

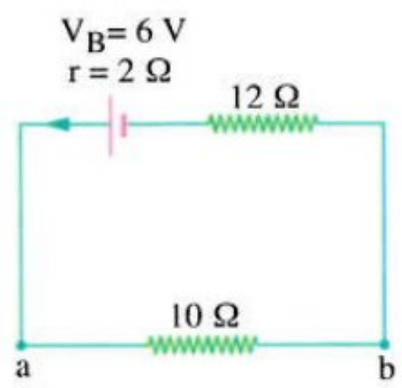
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19 **Write the mathematical relation which is represented by each of the following graphs and the equivalent for slope of the straight line in each graph:**

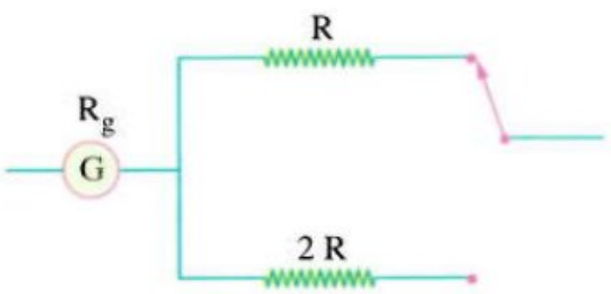


(Where: (θ) is the deflection angle of the galvanometer, (I) is the electric current intensity, (V) is the potential difference, (V_g) is the maximum potential difference at the ends of the galvanometer, (R_m) is the multiplier resistance, (R_s) is the shunt resistance, (I_g) is the maximum current intensity that can be withstood by the galvanometer).

20 In the opposite electric circuit, **compare between** the total current intensity and also the potential difference across the 10 Ω resistance before and after connecting a voltmeter of resistance 190 Ω across the points a, b. **Explain your answer.**



21 The opposite figure represents a voltmeter that consists of a galvanometer of resistance R_g, two multiplier resistors of resistances R, 2 R and a two way switch that can connect either of them to the galvanometer. **Show which of the two multiplier resistors gives the greater:**



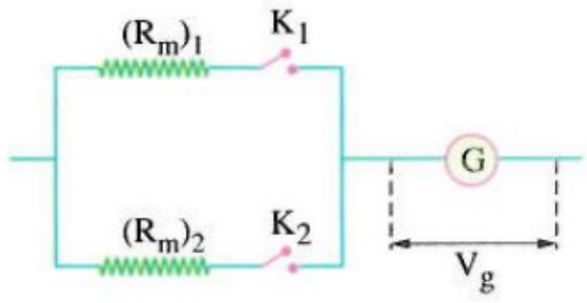
(a) Measuring range.

(b) Accuracy in measuring the potential difference.

(c) Sensitivity.

22 In the opposite figure, a galvanometer can measure a potential difference of maximum value V_g. When switch (K_1) only is closed, the device can measure a potential difference up to 7 V_g and when closing the two switches (K_1), (K_2) together, it can measure a potential difference up to 3 V_g

Determine the ratio between the two resistances $\left(\frac{(R_m)_1}{(R_m)_2}\right)$.



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The ohmmeter

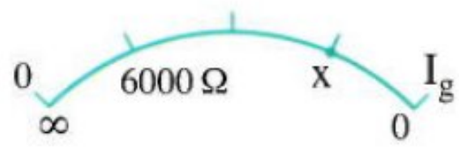
23 **In the ohmmeter:**

(a) **Why** must the emf of the electric cell be constant?

(b) When the standard value of emf of the electric cell gets decreased, **show with explanation** what is noticed when the ends of the ohmmeter are connected together?

24 **Compare with explanation between** the scales of the microammeter and the ohmmeter in terms of the regularity of the scale.

25 The opposite figure represents equal divisions on the scale of an ohmmeter that consists of a galvanometer, a standard resistance and an electric cell of emf 2 V with negligible internal resistance. **Calculate:**



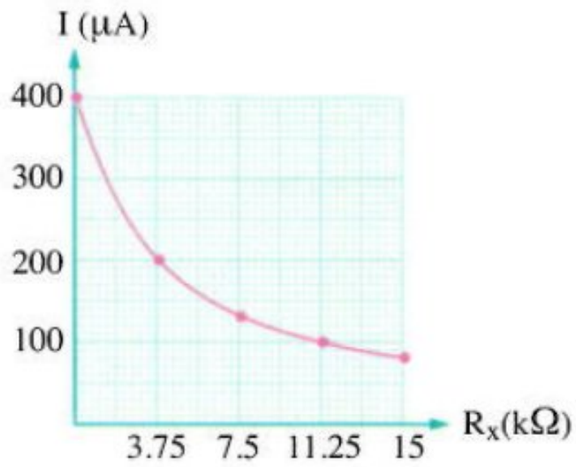
(a) The maximum reading on the scale of the galvanometer (I_g).

(b) The resistance of the external resistor that when connected to the ends of the ohmmeter, its pointer deflects to mark (x) on its scale.

26 An ohmmeter of resistance R_o contains an electric cell of emf V_B. When its ends are connected to a resistance R_x, a current of intensity I passes in its circuit.

Prove that: $R_x = \frac{V_B}{I} - R_o$

27 The opposite graph represents the relation between the electric current intensity (I) which is passing in the ohmmeter and the external resistance (R_x) which is connected to its ends. **Determine:**



(a) The maximum current that can pass through the ohmmeter.

(b) The ohmmeter resistance.

(c) The resistance of the resistor R_x which when connected to the ohmmeter, a current of 50 μA passes through it.

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Questions of the Previous Years Exams ? on Lesson 4

The moving coil galvanometer

1 In two galvanometers, if a current (I) passes therefore the pointer of the first device deflects by an angle 30° and the second deflects by 10° more than that of the first. So, by passing a current of (2 I), which of the following statements is correct after increasing the current to (2 I) in both galvanometers? *(1st session 24)*

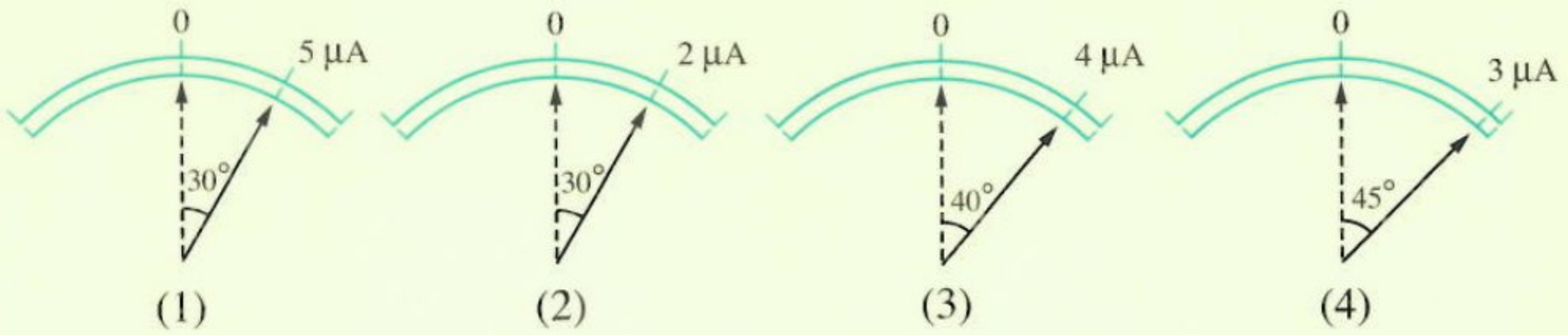
(a) The angle of deflection of the first galvanometer from zero equals 20°

(b) The sensitivity of the first galvanometer is $\frac{60}{I}$

(c) The sensitivity of the second galvanometer is $\frac{40}{I}$

(d) The angle of deflection of the second galvanometer from zero equals 40°

2 The figures show the deflection of the pointers in four galvanometers by passing different values of electric current. *(2nd session 24)*

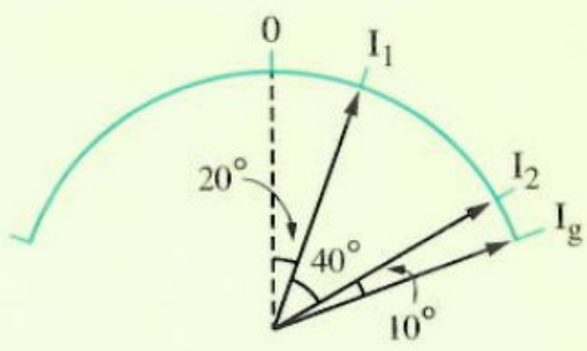


Which two galvanometers have the same sensitivity?

(a) (1), (3) (b) (1), (4) (c) (2), (4) (d) (3), (4)

3 The opposite figure represents a schematic diagram of the deflection angles of the pointer in a moving-coil galvanometer in two direct current circuits carrying two currents of intensities I_1 and I_2, respectively.

The ratio of $\left(\frac{I_1}{I_2}\right)$ is equal to ..... . *(Practice 25)*



(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{1}$ (d) $\frac{3}{1}$

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4 You have an old sensitive galvanometer in the lab with a full-scale reading of 200 μA. When a current of 50 μA passes through its coil, the pointer indicates 60 μA . Which of the following is most likely to have happened to the galvanometer as a result of frequent use? *(Practice 25)*

(a) Decrease in the magnetic permeability of the iron cylinder.

(b) Damage of the spring coils.

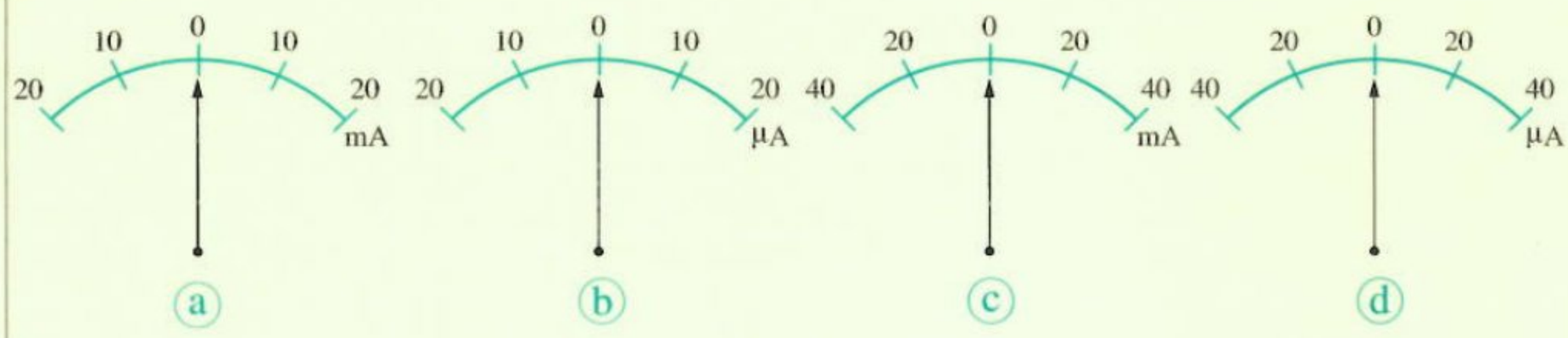
(c) The friction with agate base increases.

(d) The weakness of the magnetic field strength.

5 A moving-coil galvanometer where each 5 divisions indicate a current of 1 mA, and each 20 divisions indicate a voltage of 1V. The resistance of the galvanometer is equal to ..... . *(Practice 25)*

(a) 250 Ω (b) 500 Ω (c) 40 Ω (d) 25 Ω

6 Which of the following diagrams shows the scale of the most sensitive galvanometer, knowing that the angle between the ends of each scale is equal? *(1st session 25)*



7 Two galvanometers (X and Y) are connected in series in an electric circuit. When an electric current passes through the circuit, the pointer deflection of galvanometer (X) is observed to be greater than the pointer deflection of galvanometer (Y).

Knowing that the coils of the two devices are identical, which of the following statements explains this observation? *(2nd session 25)*

(a) In galvanometer (X), the magnetic dipole moment of the coil is greater.

(b) In galvanometer (X), friction of the coil axis with the base is greater.

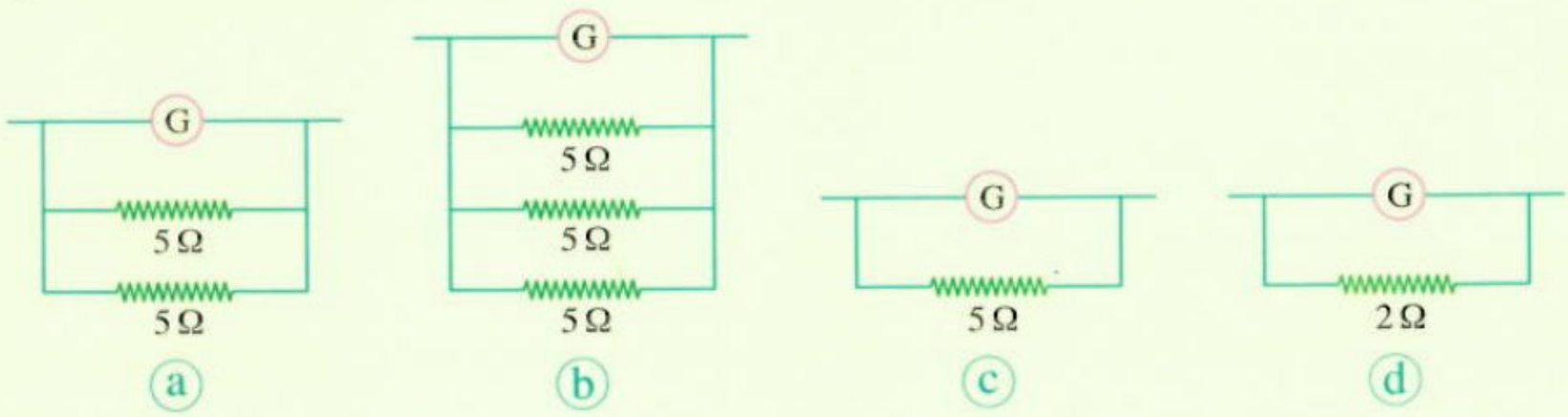
(c) In galvanometer (X), the current intensity through its coil is greater.

(d) In galvanometer (X), the magnet has more magnetic field strength.

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Ammeter

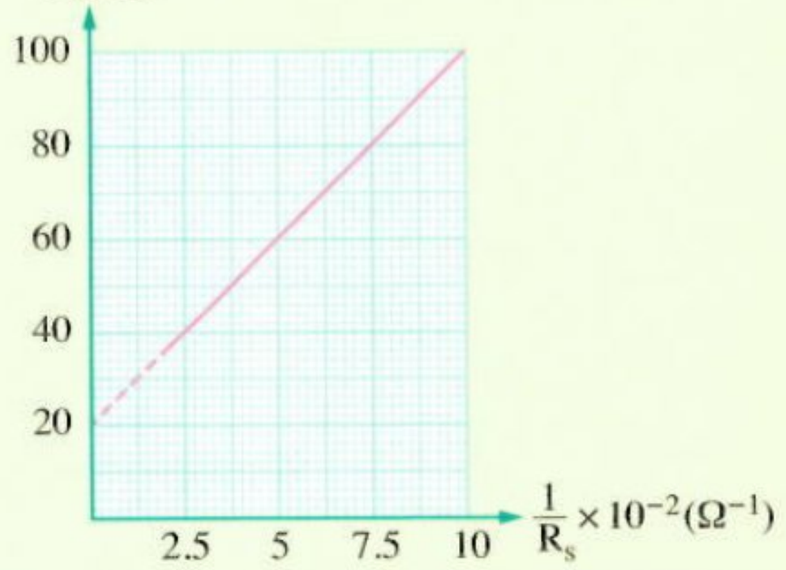
8 A sensitive galvanometer of resistance 15 Ω was connected to different shunts to convert it into an ammeter of different range in each time. Which of the following figures represents the ammeter that has the maximum range scale? *(Exp. May 21)*



9 The graph represents the relation between the maximum current intensity measured by an ammeter and the reciprocal of its shunt resistor, so the maximum potential difference between the shunt terminals equals ..... . *(Exp. June 21)*

(a) 0.1 V (b) 0.8 V

(c) 1.2 V (d) 1 V



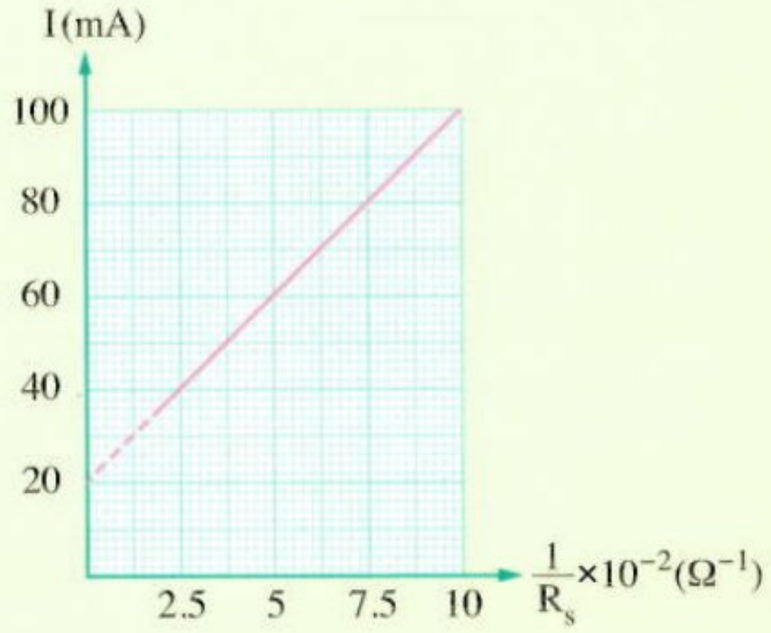
10 The opposite graph represents the relation between the maximum current intensity that could be measured by an ammeter and the reciprocal of its shunt resistor, so the value of the galvanometer resistance R_g equals .....

(a) 20 Ω *(1st session 21)*

(b) 40 Ω

(c) 80 Ω

(d) 100 Ω



11 A galvanometer, the resistance of its coil is (R_g), measures a maximum current of I_g. When a shunt resistance of value R_1 is connected to its coil, its sensitivity gets reduced to $\frac{3}{4}$ of its original value. When the shunt resistance R_1 is replaced by another of resistance R_2, its sensitivity decreases to $\frac{3}{8}$ of its original value. So, the ratio:

$\frac{\text{value of shunt } R_1}{\text{value of shunt } R_2}$ equals .....

(a) 2 (b) 3 (c) 4 (d) 5 *(2nd session 21)*

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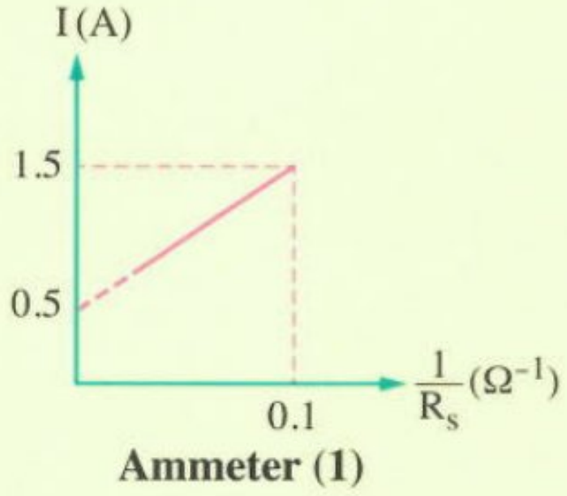
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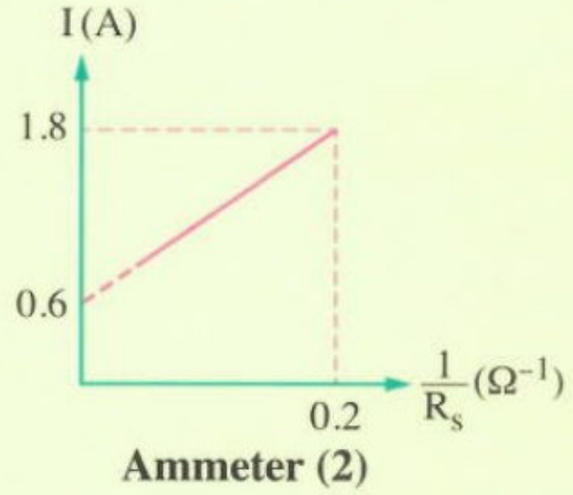
12 A galvanometer's coil has resistance (R_g) in which a current of (I_g) causes full scale deflection. When a shunt of value (R) is used, it measures a current up to ($4\ I_g$). If the shunt resistor is replaced by another of value (3 R), so the maximum current that can be measured equals ..... *(1st session 22)*

(a) $1.5\ I_g$ (b) $3\ I_g$ (c) $2.5\ I_g$ (d) $2\ I_g$

13 The graphs illustrate the relation between maximum current intensities that can be measured by two different ammeters and the reciprocal of shunt resistance $\left(\frac{1}{R_s}\right)$ in each of them, so the ratio between galvanometer's resistance in the first ammeter to that in the second one $\left(\frac{(R_g)_1}{(R_g)_2}\right)$ equals ..... . *(2nd session 22)*



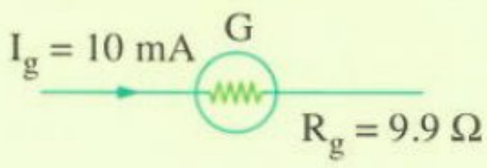
Ammeter (1)

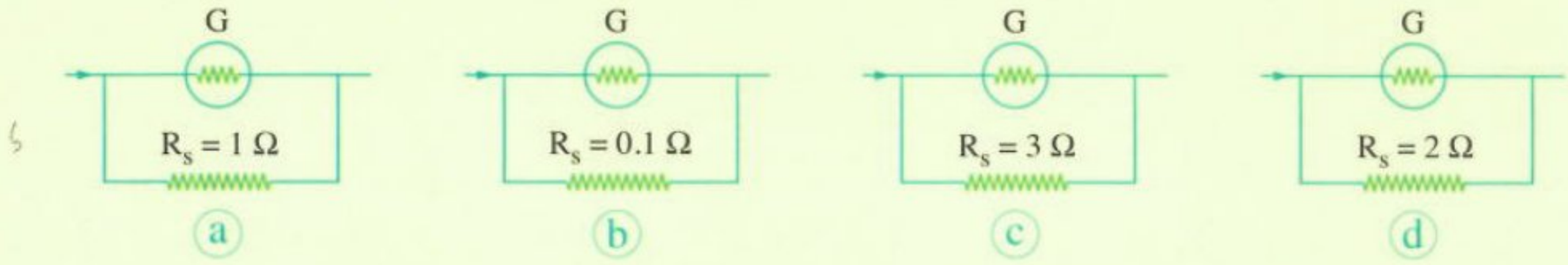


Ammeter (2)

(a) $\frac{1}{3}$ (b) $\frac{2}{1}$ (c) $\frac{3}{1}$ (d) $\frac{1}{2}$

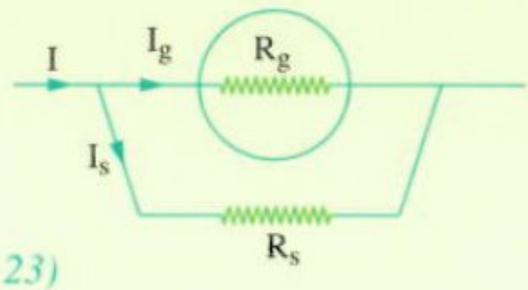
14 The figure represents a sensitive galvanometer.
Which of the following figures correctly represents the modification to the galvanometer to become capable of measuring a maximum current of 1A? *(Exp. 23)*





(a) (b) (c) (d)

15 In the opposite figure:
If the shunt is replaced to increase the sensitivity of the device keeping the current (I) constant, which of the ratios below increases? *(1st session 23)*



(a) $\frac{I_g}{I_s}$ (b) $\frac{V_g}{V_s}$ (c) $\frac{R_g}{R_{eq}}$ (d) $\frac{R_g}{R_s}$

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16 The opposite table represents different values of shunt resistances R_s that are used one at a time with the same galvanometer to form direct current ammeters W, X, Y and Z, so what is the correct order of these ammeters according to their sensitivities? *(2nd session 23)*

	R_S
W	20 Ω
X	5 Ω
Y	40 Ω
Z	10 Ω

(a) Y > W > X > Z (b) X > Z > W > Y

(c) Y > W > Z > X (d) W > Y > Z > X

17 A galvanometer with coil resistance (R_g) is connected to a shunt resistance of value ($\frac{1}{2}$ R_g), then the galvanometer is reconnected to another shunt resistance of value ($\frac{1}{4}$ R_g).

So, the ratio: $\frac{\text{Sensitivity of ammeter in first case}}{\text{Sensitivity of ammeter in second case}}$ = ... *(1st session 24)*

(a) $\frac{1}{5}$ (b) $\frac{3}{5}$ (c) $\frac{1}{3}$ (d) $\frac{5}{3}$

18 A galvanometer, the resistance of its coil equals 60 Ω , so the value of the shunt resistor that decreases its sensitivity to $\frac{1}{6}$ its value = ... *(2nd session 24)*

(a) 24 Ω (b) 6 Ω (c) 3 Ω (d) 12 Ω

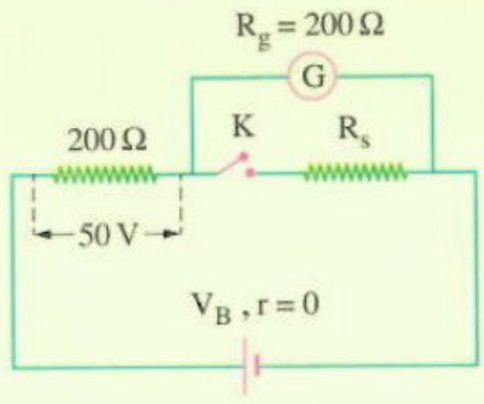
19 The resistance of a microammeter coil is 100 Ω and the maximum current that its coil can measure is 100 μA. So, its range can be increased to measure electric currents up to 0.1 A by connecting it to a shunt resistor of resistance ..... . *(Practice 25)*

(a) 0.5 Ω (b) 0.1 Ω (c) 0.05 Ω (d) 0.01 Ω

20 A galvanometer with a coil resistance of 100 Ω has a scale divided into number of divisions (N). When a current of intensity (I) passes through it, its pointer deflects to the end of the scale. When a 20 Ω resistor is connected in parallel to the galvanometer to convert it into an ammeter, and the same current (I) is passed, its pointer deflects to only two divisions. Then the number of divisions on the galvanometer scale equals ..... . *(Practice 25)*

(a) 2 (b) 6 (c) 12 (d) 24

21 In the opposite figure, if switch K is closed, the sensitivity of the galvanometer decreases to one-fifth. Then, the values of R_S and V_B respectively are ..... . *(Practice 25)*



(a) 40 Ω, 100 V (b) 100 Ω, 60 V

(c) 50 Ω, 100 V (d) 40 Ω, 60 V

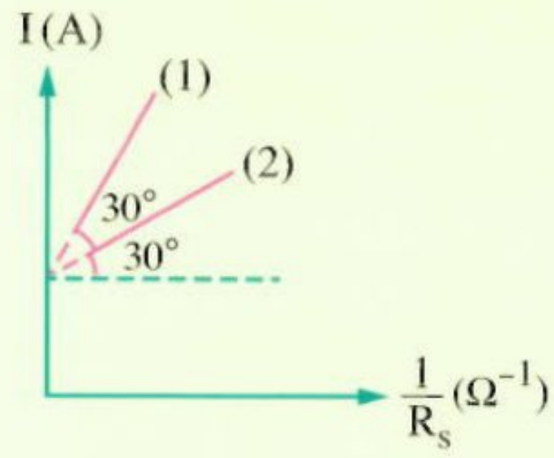
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22 The opposite graph represents the relation between the maximum measurement range of two ammeters (I) and the reciprocal of its shunt resistor $\left(\frac{1}{R_s}\right)$ which connected to the galvanometer coil in each of them. The ratio between the resistances of the two galvanometers $\left(\frac{(R_g)_1}{(R_g)_2}\right)$ equals ..... .

(Practice 25)



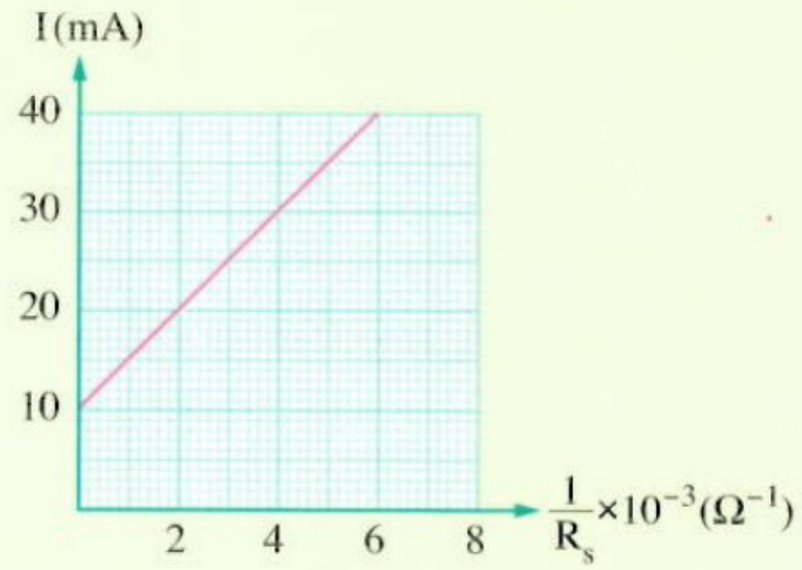
(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{3}$ (d) $\frac{3}{1}$

23 An ammeter of resistance (R) consists of a galvanometer of resistance 45 Ω and a shunt resistance (R_s). If the current flowing through the galvanometer is $\frac{1}{5}$ of the current flowing through the shunt, what is the resistance (R) of the ammeter?

(1st session 25)

(a) 9 Ω (b) 7.5 Ω (c) 10 Ω (d) 15 Ω

24 An ammeter contains a galvanometer of resistance (R_g) and with full-scale reading (I_g). A shunt resistance (R_s), whose value can be varied, is connected in parallel to the galvanometer. The graph represents the relationship between the maximum electric current intensity (I) that can be measured by the ammeter and the reciprocal of the shunt resistance $\left(\frac{1}{R_s}\right)$. What is the potential difference across the galvanometer when its pointer deflects to full-scale?

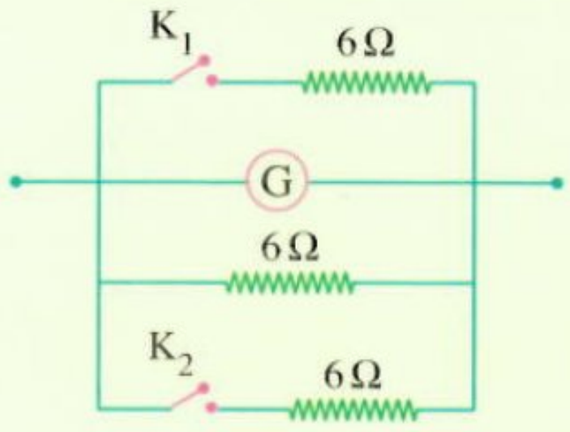


(2nd session 25)

(a) 0.5 V (b) 5000 V (c) 5 V (d) 5×10^{-3} V

25 The figure represents a galvanometer (G) converted into an ammeter. If you know that the resistance of the galvanometer's coil is $R_g = 18\ \Omega$, then the ratio between the sensitivity of the device before and after closing both switches K_1 and K_2 together is equal to ..... .

(Practice 26)

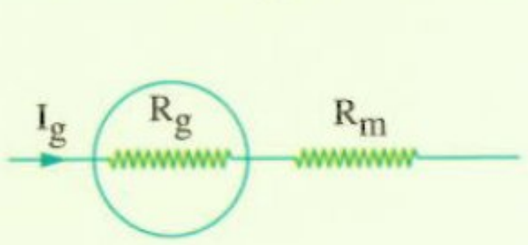


(a) $\frac{2}{5}$ (b) $\frac{5}{2}$ (c) $\frac{7}{4}$ (d) $\frac{4}{7}$

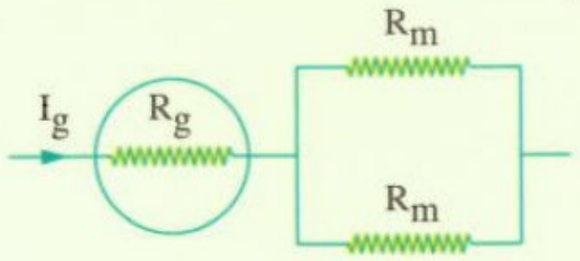
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The voltmeter

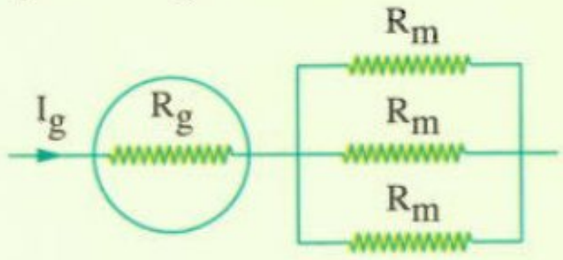
26 Three galvanometers of equal coil resistances R_g were connected to three multipliers to convert them into three voltmeters (A, B and C) as shown in the following figures, so the correct arrangement for the maximum readings of them (V_A, V_B and V_C) will be .....



Voltmeter (A)



Voltmeter (B)



Voltmeter (C)

(a) $V_B > V_A > V_C$ (b) $V_A < V_C < V_B$

(c) $V_C > V_B > V_A$ (d) $V_C < V_B < V_A$ *(Exp. June 21)*

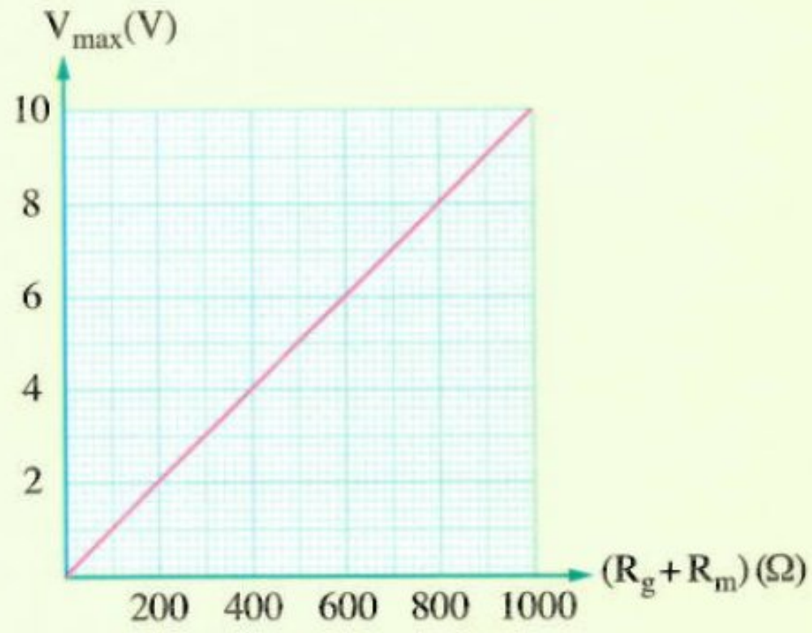
27 A galvanometer that has a coil of 50 Ω resistance was connected to a multiplier of 450 Ω, hence its maximum reading was 1 V and when it was connected to another multiplier (R_{m2}), its maximum reading became 18 V, the value of R_{m2} = ... *(1st session 21)*

(a) 9000 Ω (b) 8950 Ω (c) 9050 Ω (d) 9500 Ω

28 A galvanometer measures 0.1 volt of maximum potential difference when a maximum current of 2 mA passes in it, each division of its scale indicates 0.01 V, if it is connected to a multiplier resistance of 450 Ω, each division will indicate ..... . *(2nd session 21)*

(a) 0.01 V (b) 1 V (c) 0.1 V (d) 0.001 V

29 A galvanometer that can withstand maximum potential difference of 1 volt across its coil, was connected to several multipliers to convert it into voltmeter.
The graph illustrates the maximum measured potential difference (V_{max}) versus the total resistance of the voltmeter ($R_g + R_m$). So, the resistance of the galvanometer's coil (R_g) equals ..... . *(1st session 22)*



(a) 100 Ω (b) 1000 Ω (c) 500 Ω (d) 50 Ω

30 A voltmeter, its resistance is 100 Ω and the maximum voltage that can be measured by it is 1 V. The value of multiplier resistance that should be connected to increase the measured voltage by 10 times equals ..... . *(2nd session 22)*

(a) 0.9 kΩ (b) 10 kΩ (c) 1.1 kΩ (d) 1 kΩ

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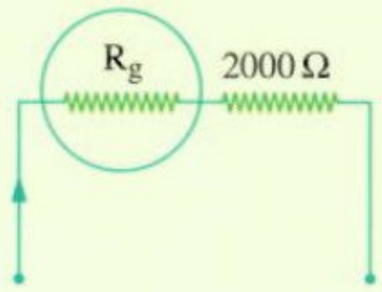
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31 A voltmeter, the resistance of its coil is 40 Ω, reaches its full-scale deflection when a current of intensity 0.1 A passes through it. Then the value of the multiplier resistance of the device that makes the maximum potential difference reach 100 V between its two ends is ..... .

(Exp. 23)

(a) 25 Ω (b) 2.5 Ω (c) 960 Ω (d) 1040 Ω

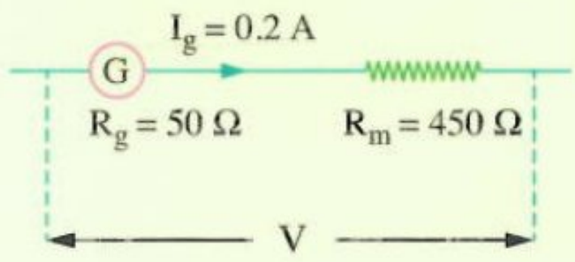
32 A galvanometer of maximum reading 0.01 A is connected in series to a resistor of 2000 Ω to convert it into voltmeter as shown in the figure so that the maximum potential difference that can be measured by the voltmeter was 20.5 V. In order to measure a maximum potential difference 10.25 V, the 2000 Ω resistor must be replaced by a resistor of resistance equal to ..... .



(1st session 23)

(a) 1025 Ω (b) 1000 Ω (c) 975 Ω (d) 4000 Ω

33 According to the data shown in the figure, the potential difference measured by the voltmeter = ...



(a) 50 V (b) 100 V
(c) 20 V (d) 10 V *(2nd session 23)*

34 A voltmeter of total resistance 1200 Ω and the maximum voltage that its coil can withstand is 3 V, if a voltage multiplier (R_m) is connected to its coil, the maximum voltage that the device can withstand is 10 V, then the value of the voltage multiplier (R_m) is equal to ..... . *(Practice 25)*

(a) 1800 Ω (b) 2400 Ω (c) 2800 Ω (d) 4000 Ω

35 A moving coil galvanometer with coil resistance R_g can withstand a maximum voltage (V_g) it can withstand. When it is connected in series to a resistance R to convert it into a voltmeter with range V_1. And when the same galvanometer is connected in series to a resistance of 2 R to convert it into a voltmeter with range of V_2, then ..... *(Practice 25)*

(a) $V_2 = V_1 - 2V_g$ (b) $V_2 = 2V_1 - V_g$ (c) $V_2 = 2V_1 + V_g$ (d) $V_2 = 2V_1$

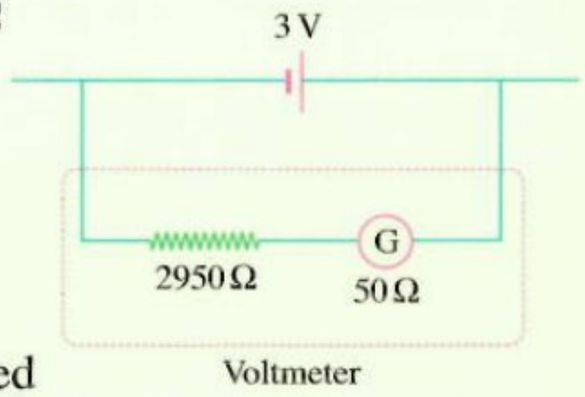
36 A moving coil galvanometer has 15 divisions. If every 5 divisions represent 5 mA, and every 2 divisions represent 10 mV, the value of resistance that should be connected in series to the galvanometer coil so that each one division represents 1 V is ..... .

(Practice 25)

(a) 995 Ω (b) 99.5 Ω (c) 9.95 Ω (d) 9950 Ω

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37 ✳ A voltmeter consists of a galvanometer of resistance 50 Ω and a multiplier of resistance 2950 Ω. When the voltmeter is connected to a battery with an electromotive force of 3 V and negligible internal resistance as shown in the figure, its pointer deflects to its full scale. While the voltmeter is connected to the poles of the battery, a resistor (R) is connected to the multiplier inside it, causing its pointer to deflect to half scale. What is the value of resistance (R) and its way of connection? *(1st session 25)*



(a) 1500 Ω, in series.
(b) 3000 Ω, in parallel.
(c) 1500 Ω, in parallel.
(d) 3000 Ω, in series.

Ohmmeter

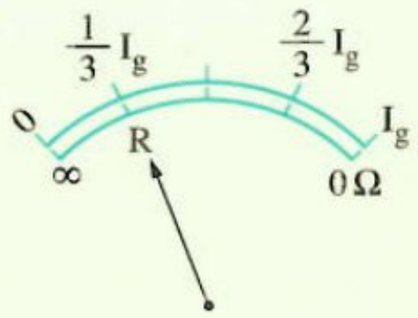
38 An ohmmeter was connected with an external resistor (x) of resistance 400 Ω, so its pointer deflected to $\frac{3}{4}$ of the galvanometer scale. If (x) is replaced with another resistor (y) of 6000 Ω, the pointer deflects to ..... of the galvanometer scale. *(Exp. May 21)*

(a) $\frac{1}{6}$ (b) $\frac{3}{5}$ (c) $\frac{1}{5}$ (d) $\frac{5}{6}$

39 An ohmmeter contains a galvanometer that shows full-scale deflection when a current I_g passes in it and when an external resistor 12 kΩ is connected to its test terminals, the passing current becomes $\frac{1}{5} I_g$, so if a resistor of 1.5 kΩ is connected to the test terminals, the passing current becomes ..... . *(Exp. June 21)*

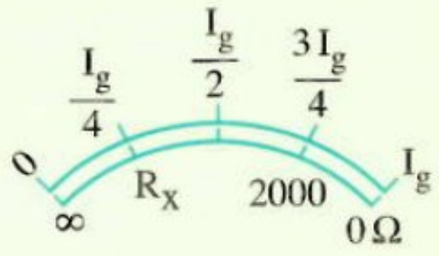
(a) $\frac{2}{3} I_g$ (b) $\frac{1}{8} I_g$ (c) $\frac{1}{5} I_g$ (d) $\frac{3}{4} I_g$

40 The opposite figure represents the scale of a galvanometer in an ohmmeter, when it is connected with external resistance R, the pointer has deflected to $\frac{1}{3} I_g$, so the resistance of the device equals ..... .



(a) 0.5 R (b) R
(c) 2 R (d) 3 R *(1st session 21)*

41 The opposite figure represents equal divisions on the scale of the galvanometer in the ohmmeter circuit, so the value of R_x equals ..... .



(a) 6000 Ω (b) 18000 Ω
(c) 12000 Ω (d) 10000 Ω *(2nd session 21)*

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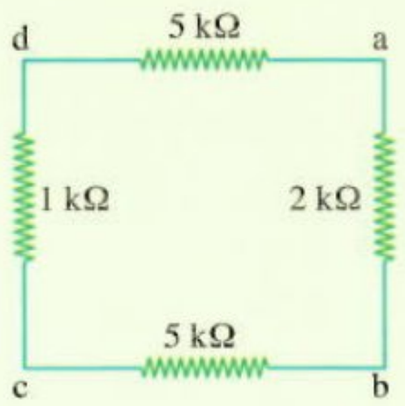
42 An ohmmeter contains a galvanometer that shows a full-scale deflection when a current I_g passes in it, when an external resistor 50 kΩ is connected to its two terminals, the passing current becomes $\frac{1}{3} I_g$, so the external resistance that makes the current passing in the ohmmeter become $\frac{3}{4} I_g$ equals ..... . *(1st session 22)*

(a) $\frac{25}{3}$ kΩ (b) $\frac{225}{2}$ kΩ (c) $\frac{50}{3}$ kΩ (d) $\frac{50}{4}$ kΩ

43 An ohmmeter contains a galvanometer of current full-scale deflection I_g. When an external resistor (R) is connected to its two terminals, the passing current becomes $\frac{3}{4} I_g$, then when the resistor (R) is replaced with another resistor of value (3 R), the passing current becomes ..... . *(2nd session 22)*

(a) $\frac{1}{4} I_g$ (b) $\frac{1}{3} I_g$ (c) $\frac{4}{9} I_g$ (d) $\frac{1}{2} I_g$

44 Four electric resistances are connected as in the figure, the pointer of ohmmeter reads the same reading when the device is connected between ..... . *(Exp. 23)*



(a) the two points (c), (b) or the two points (b), (d)

(b) the two points (a), (c) or the two points (a), (d)

(c) the two points (a), (c) or the two points (b), (d)

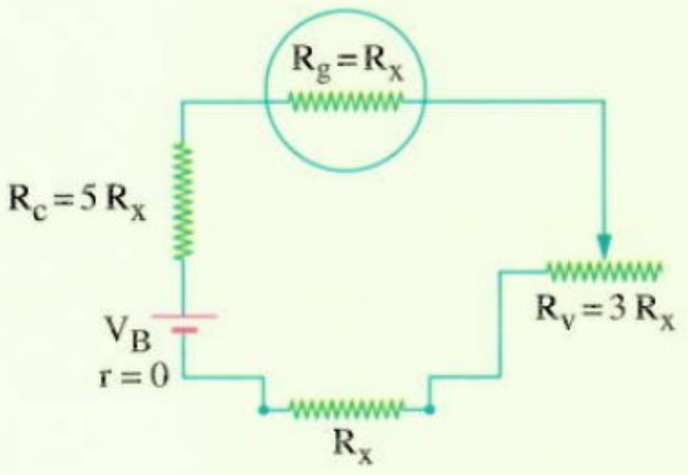
(d) the two points (a), (d) or the two points (c), (d)

45 Ohmmeter of internal resistance 3000 Ω, its pointer deflects by an angle (θ) when its terminals are connected together and when it is connected with resistance R_1, the pointer deflects by an angle $(\frac{\theta}{3})$ and by replacing R_1 by another resistance R_2, the pointer deflects by an angle $(\frac{\theta}{4})$, so the values of R_1 and R_2 respectively equal ..... . *(Exp. 23)*

(a) 3000 Ω, 9000 Ω (b) 6000 Ω, 12000 Ω

(c) 3000 Ω, 12000 Ω (d) 6000 Ω, 9000 Ω

46 In the ohmmeter circuit shown in the figure, when another resistance is connected to the unknown resistance (R_x) in series, the pointer deflects to $\frac{3}{5}$ from the scale of the galvanometer. Then the value of the connected resistance is ..... . *(1st session 23)*

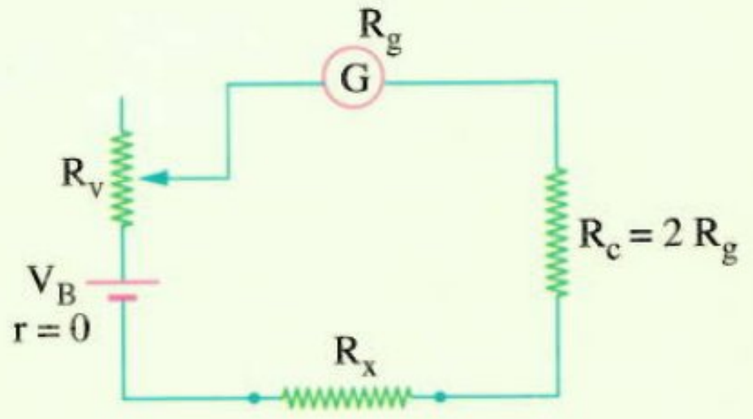


(a) $6 R_x$ (b) $5 R_x$

(c) $\frac{2}{3} R_x$ (d) $3 R_x$

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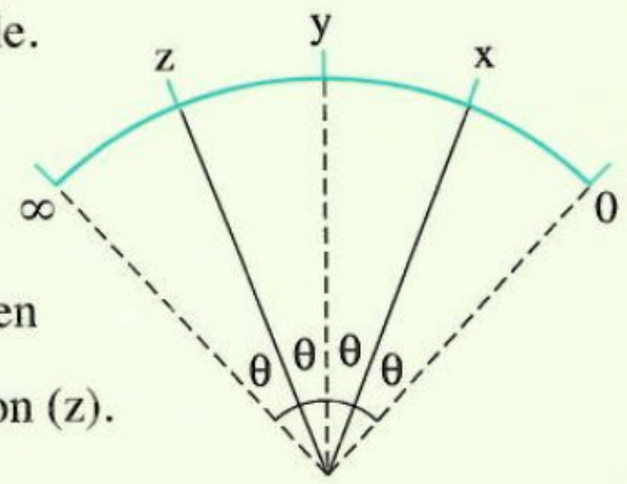
47 The opposite figure represents an ohmmeter circuit containing a galvanometer that the resistance of its coil is (R_g), when it is connected to an external resistance (R_x) that equals (15 R_g) the pointer of the galvanometer deflects to $\frac{1}{5}$ of its full current scale deflection. Therefore the value of the resistance taken from the rheostat (R_v) = ...



(2nd session 23)

(a) 3.75 R_g
(b) 0.75 R_g
(c) 0.25 R_g
(d) 3.25 R_g

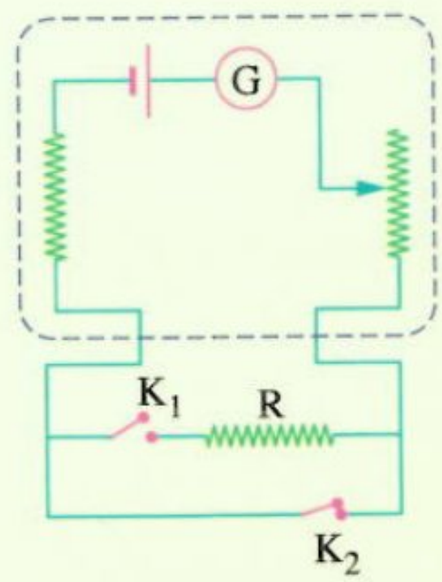
48 ✱ The diagram shows equal spacings on an ohmmeter scale. When a resistor of value (R) is connected between the ohmmeter terminals, its pointer deflects to position (x). When another resistor is connected with resistor (R) between the terminals of the ohmmeter, its pointer deflects to position (z). The value of the resistance connected to resistor R is equal to ..... .



(a) 9 R
(b) 2 R
(c) 8 R
(d) 3 R

(1st session 25)

49 ✱ An ohmmeter has resistance R and its terminals are connected to an external resistance R and two switches, as shown in the figure. The ohmmeter pointer stands at reading (∞) on the resistance scale when ..... .



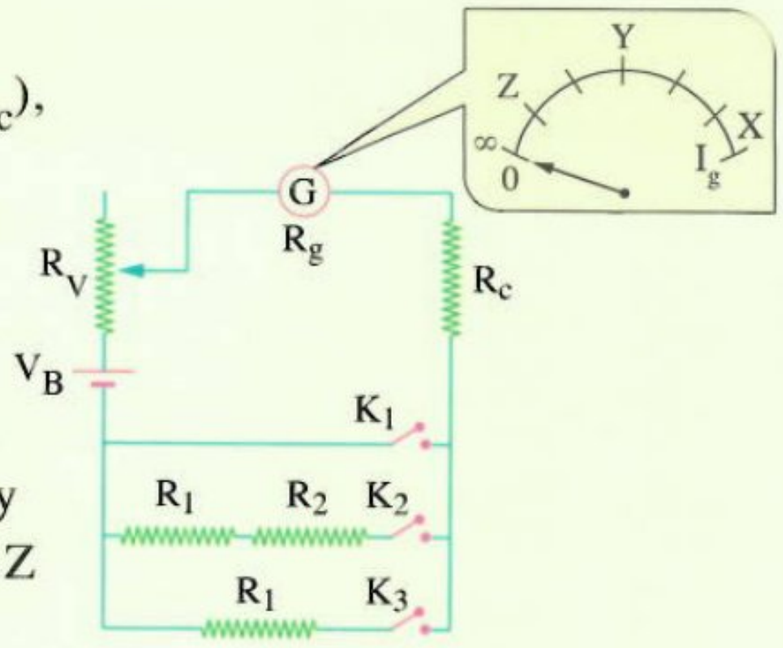
(a) both switches are closed
(b) both switches are open
(c) switch K_1 is open, and switch K_2 is closed
(d) switch K_1 is closed, and switch K_2 is open

(2nd session 25)

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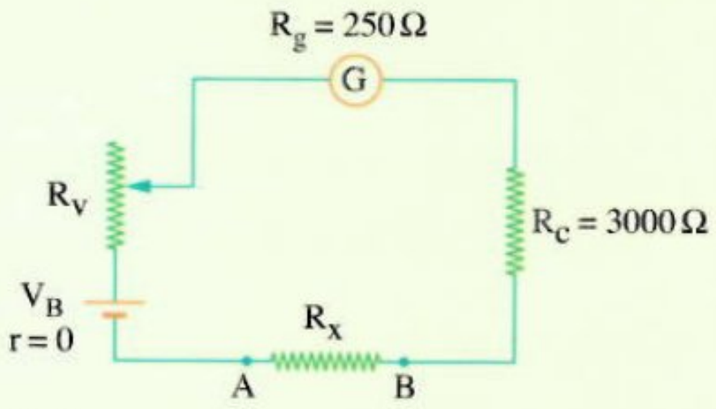
50 The figure represents an ohmmeter consisting of a galvanometer of resistance (R_g), a fixed resistor (R_c), a variable resistor (R_v), a battery of electromotive force (V_B) and negligible internal resistance and a device scale divided into equal divisions. The ohmmeter is connected to several switches and several unknown resistors. When only switch K_1 is closed, the pointer deflects to position X, when only switch K_2 is closed, the pointer deflects to position Z and when both switches K_2 and K_3 are closed, the pointer deflects to position Y, so the ratio $\frac{R_1}{R_2}$ is equal to ..... .



(Practice 26)

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{1}$ (d) $\frac{3}{2}$

51 The opposite figure illustrates the structure of an ohmmeter, and the maximum reading on the scale of the galvanometer is 900 μA if known that an external resistance (R_x) of 10 kΩ causes a current of intensity 300 μA to pass through the device.



Calculate:

(a) The value of the resistance taken from the rheostat (R_v).

(b) The value of emf of the ohmmeter battery (V_B).

(1st session 24)

52 An ohmmeter has a total resistance of 3750 Ω.

Calculate:

(a) The value of the external resistance R_x that makes the pointer of the ohmmeter deflect to $\frac{I_g}{3}$

(b) The value of the resistance (R) that should be connected in parallel with R_x to make the pointer deflect to $\frac{3\,I_g}{4}$

(2nd session 24)

53 An ohmmeter of resistance 1500 Ω, whose pointer of the galvanometer deflects to its full scale deflection when a current of 200 mA passes through it. At that moment, the angle of deflection of the pointer from the zero position is 80°. *(Practice 25)*

Calculate:

(a) The deflection angle of the galvanometer pointer when an external resistance of 4500 Ω is connected across the terminals of the ohmmeter.

(b) The electric current intensity passing through the galvanometer that causes the pointer to deflect by an angle of 60°.

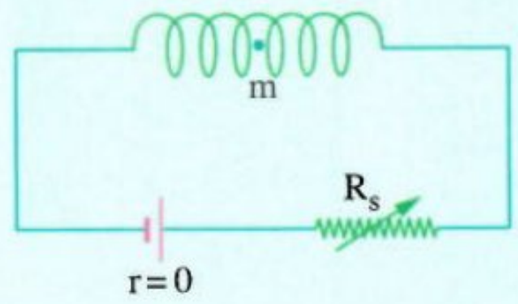
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Test on Chapter 2

Answered

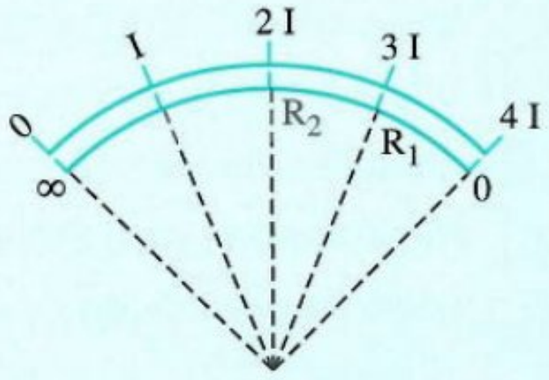
First Choose the correct answer

1 In the opposite circuit, a solenoid is connected to a variable resistance (R_s) and a battery of negligible internal resistance, if the resistance R_s is increased, the magnitude of the magnetic field due to the solenoid at its center (point m) will ..... .



(a) increase (b) decrease (c) not change (d) vanish

2 The opposite figure shows an ohmmeter scale of equal divisions, so the ratio $\left(\frac{R_1}{R_2}\right)$ is ..... .



(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $\frac{3}{2}$ (d) $\frac{1}{2}$

3 When a solenoid of length ℓ, number of turns N and cross-sectional radius r is connected to a battery of negligible internal resistance, a magnetic flux of density B is formed at the center of the solenoid, so which of the following causes the magnetic flux density at the center of the solenoid to become 3 B?

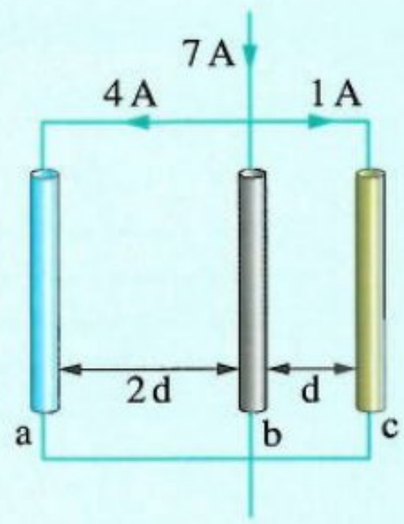
(a) Compressing the solenoid turns to decrease its length to $\frac{\ell}{3}$

(b) Increasing the length of the solenoid to 3 ℓ.

(c) Increasing the cross-sectional radius of the solenoid to 3 r while keeping ℓ and N unchanged.

(d) Cutting off one third of the solenoid and connecting the remaining part to the same battery.

4 Three straight wires a, b and c are made of different materials and they have the same dimensions. They are placed parallel in the same plane and connected to an electric circuit so that are an electric current passes through each of them as shown in the opposite figure, which of the wires are affected by a resultant magnetic force in the same direction?



(a) Wires a, b (b) Wires c, b

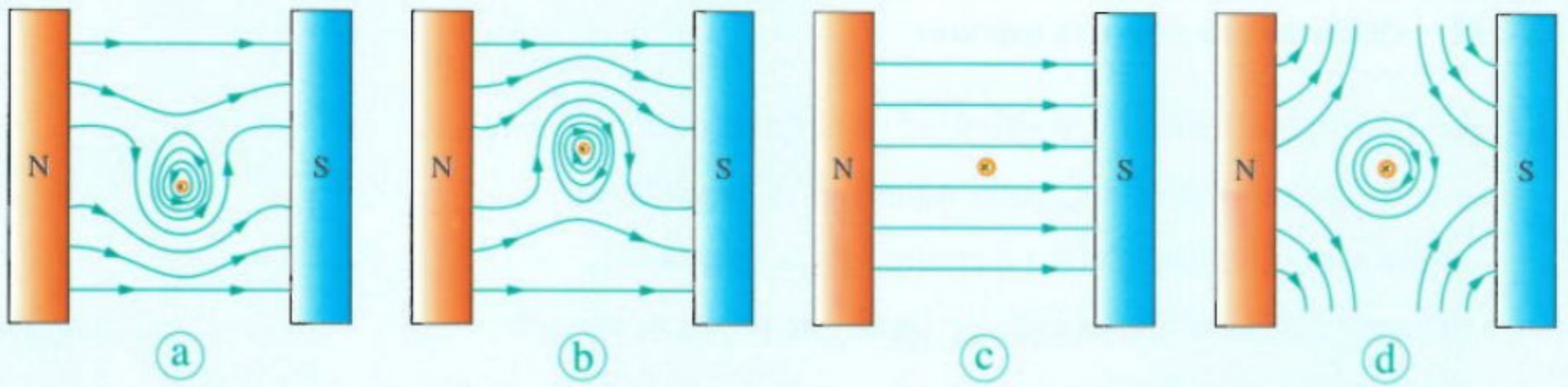
(c) Wires a, c (d) The three wires.

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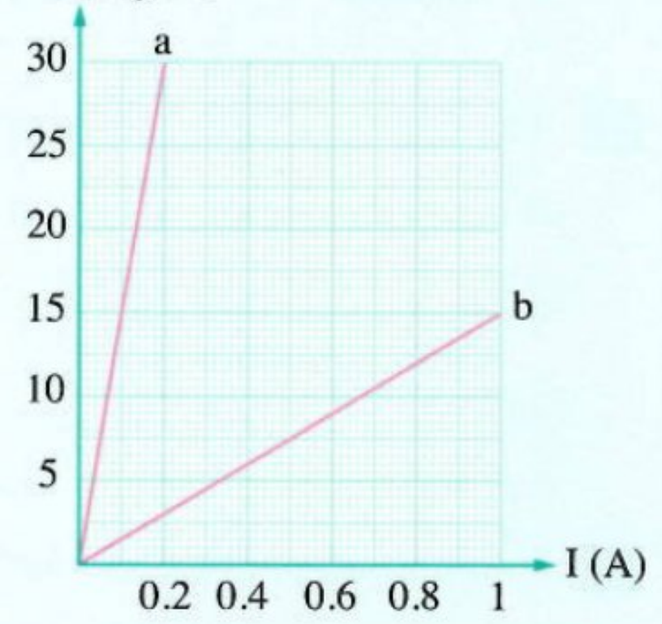
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5 Which of the following figures represents correctly the magnetic flux lines when a straight wire carrying an electric current into the page is placed between the two poles of a magnet?



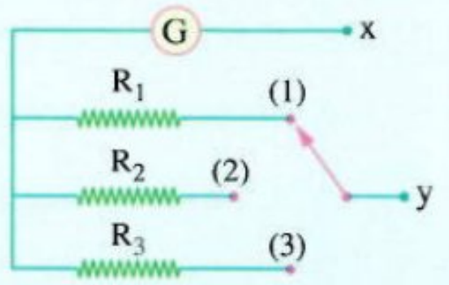
6 The two graphical lines a, b shown in the opposite figure represent the relation between the deflection angle (θ) of the galvanometer's pointer and the current intensity (I) in its circuit before and after converting the galvanometer into an ammeter, then the ratio of the ammeter's sensitivity to that of the galvanometer respectively equals ..... .

(a) 0.01 (b) 0.02
(c) 0.1 (d) 0.2

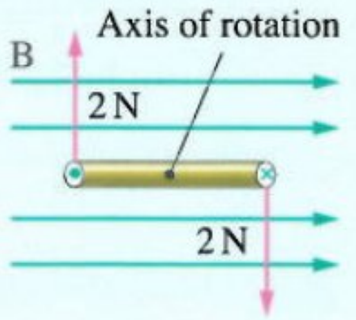


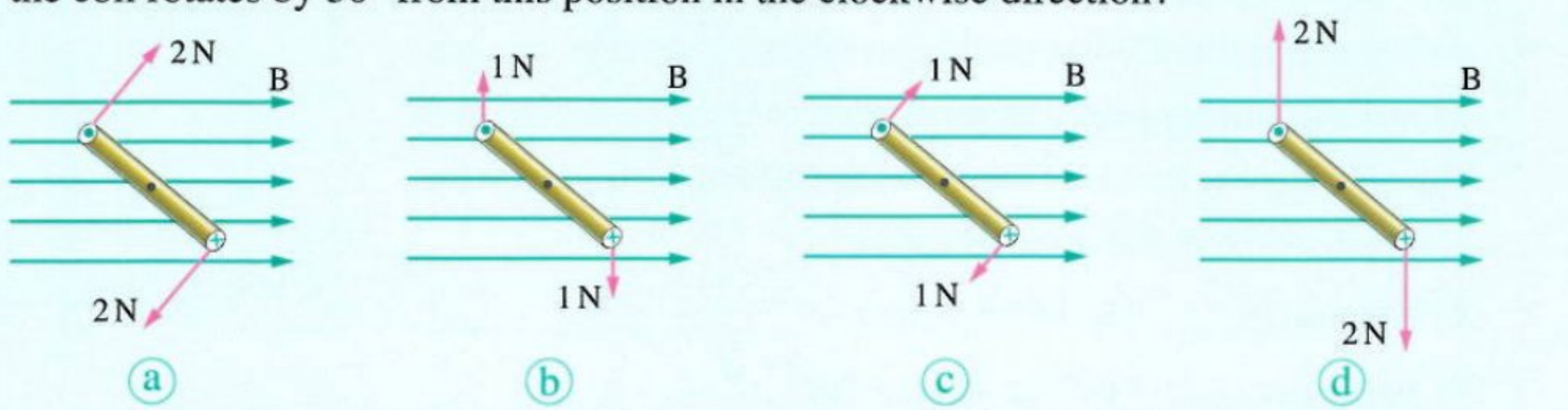
7 The opposite schematic diagram represents an electric device (xy) which is ..... .

(a) multi-scale ammeter (b) multi-scale voltmeter
(c) digital ammeter (d) ohmmeter



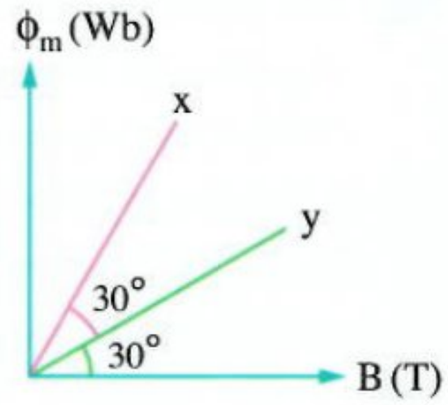
8 A rectangular coil carrying an electric current is placed parallel to a uniform magnetic field (B) and the opposite figure represents the magnitude and the direction of the two forces acting on the two sides of the coil parallel to its axis of rotation, which of the following figures represents the magnitude and the direction of these forces if the coil rotates by 30° from this position in the clockwise direction?





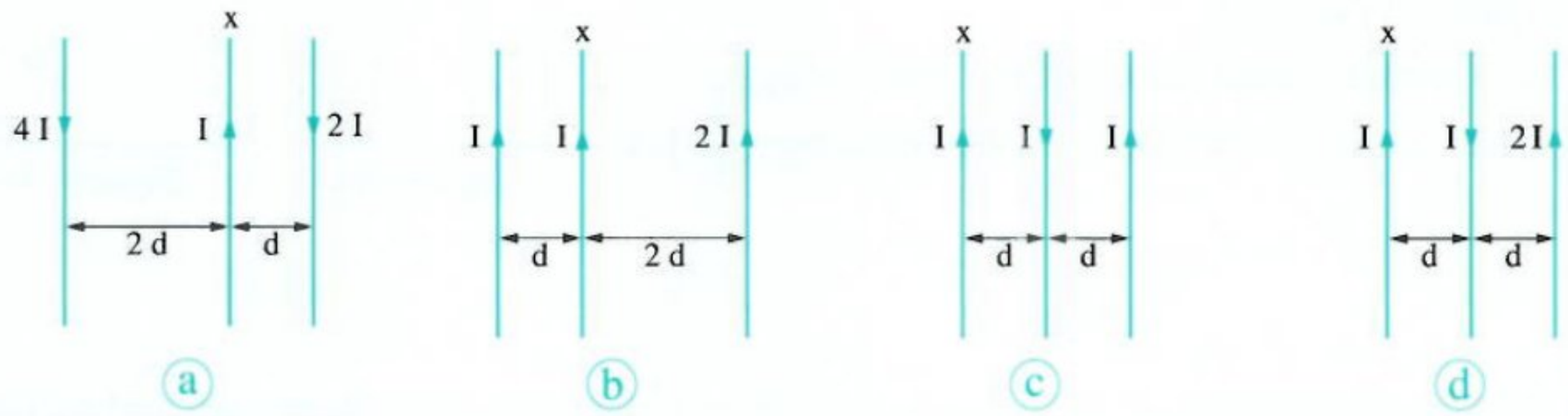
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9 Two circular rings x, y are placed perpendicular to a uniform magnetic field of changeable flux density, the opposite graph represents the relation between the magnetic flux (ϕ_m) that passes through the two rings and the magnetic flux density (B) that acts on them, then the ratio between the two diameters of the two rings $\left(\frac{d_x}{d_y}\right)$ is equal to ..... .

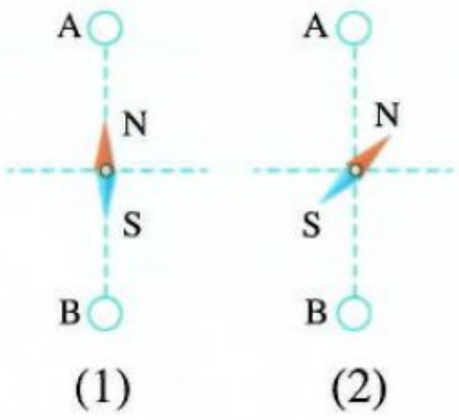


(a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{1}$

10 Each of the following figures represents the position of wire x with respect to the other two wires, if you know that the wires are parallel and in the same plane and carry electric currents, so in which of the following figures is the wire x affected by a resultant magnetic force?



11 A magnetic needle was placed in the plane of the magnetic meridian at the mid distance between two straight vertical wires (A), (B), the needle settled as in figure (1) affected by the Earth's magnetic field when no electric current passes through any of them. When an electric current of the same intensity passes through each of them, the needle deflected as in figure (2), then the directions of the currents through the two wires (A), (B) respectively are ..... .



(a) into the page, into the page

(b) into the page, out of the page

(c) out of the page, into the page

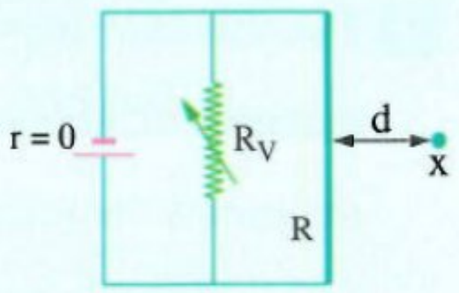
(d) out of the page, out of the page

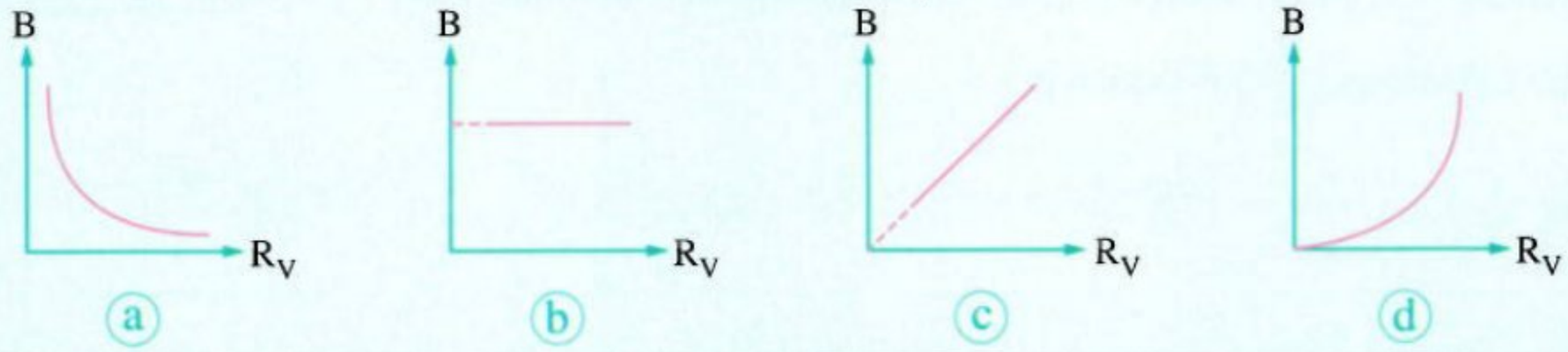
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12 The opposite figure represents an electric circuit consisting of a straight wire of resistance R, variable resistance R_v and a battery of negligible internal resistance. So, which of the following graphs represents the relation between the magnetic flux density (B) at point x due to the flow of an electric current in the straight wire and the variable resistance of R_V?





13 Two identical straight wires x, y each of resistance R are connected together and to a battery of negligible internal resistance in two different methods as in the figures (1) and (2).

So, the ratio of the values of the two magnetic forces acting on the wire x in the two cases $\left(\frac{F_1}{F_2}\right)$ is equal to ..... .

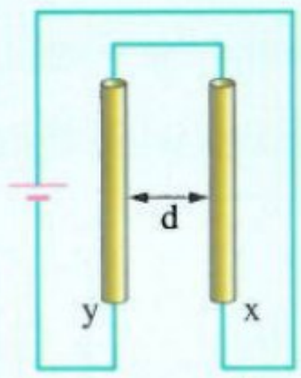


Figure (1)

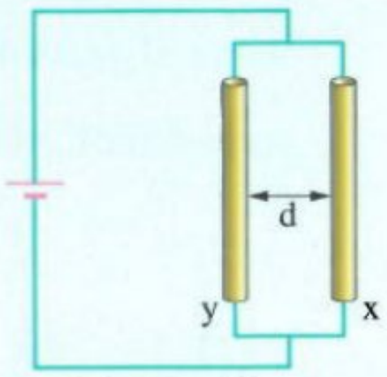


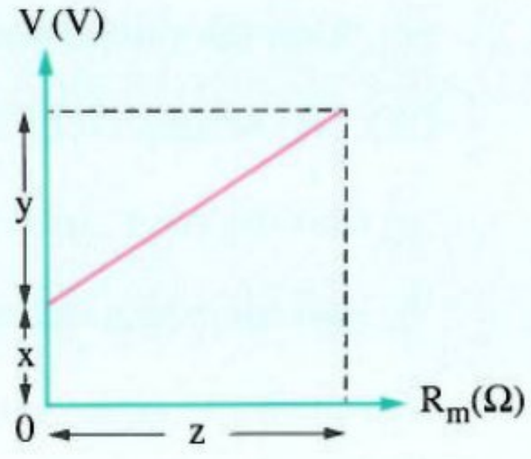
Figure (2)

ⓐ $\frac{1}{2}$ ⓑ $\frac{2}{1}$ ⓒ $\frac{1}{4}$ ⓓ $\frac{4}{1}$

14 Each of the following figures represents two circular coils, their planes are horizontal and each has the same current intensity. In which case does, the magnetic flux pattern in the shaded region between the two coils resemble the magnetic flux pattern between the poles of a U-shaped magnet where the direction of the flux lines is upward?



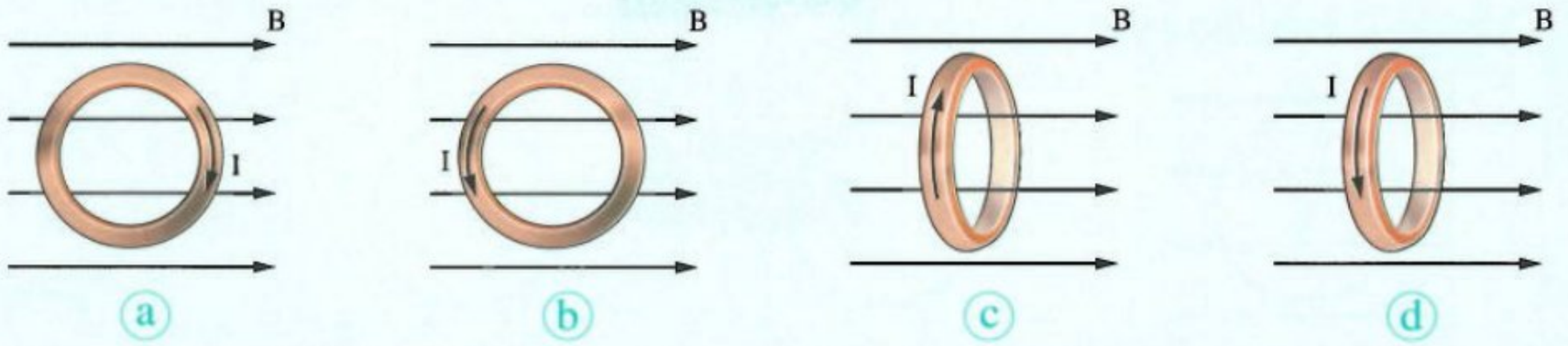
15 A voltmeter consists of a galvanometer of resistance (R_g) and a multiplier resistance whose its value can be changed, the opposite graph represents the relation between the maximum voltage (V) that can be measured by the voltmeter and the multiplier resistance (R_m) inside it, then the resistance of the galvanometer (R_g) in terms of the data shown in the graph equals ..... .



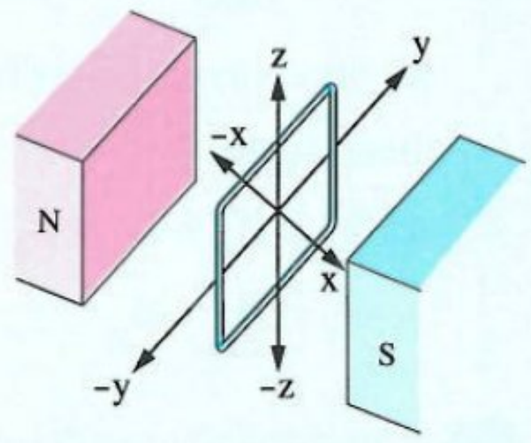
ⓐ $\frac{xz}{y}$ ⓑ $\frac{yz}{x}$ ⓒ $\frac{y}{xz}$ ⓓ $\frac{z}{xy}$

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16 A metallic ring carrying an electric current of intensity (I) is placed in a magnetic field of flux density (B), in which of the following cases is the direction of the magnetic dipole moment of the ring is parallel to the page and rightwards?

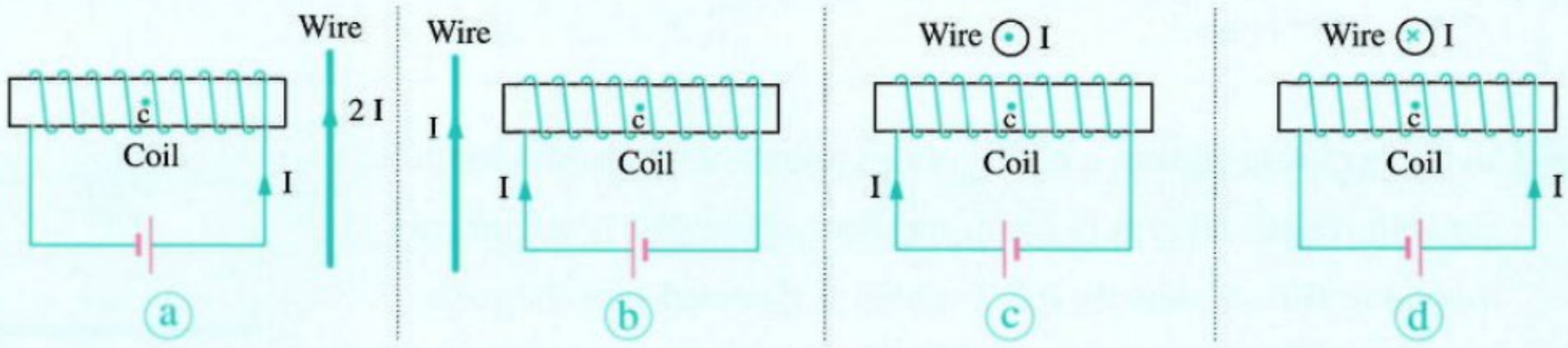


17 In the opposite figure, a coil is placed perpendicular to a magnetic field, if the magnetic flux that passes through it is ϕ_m , what procedure makes the change in the flux passing through it become $2\ \phi_m$?

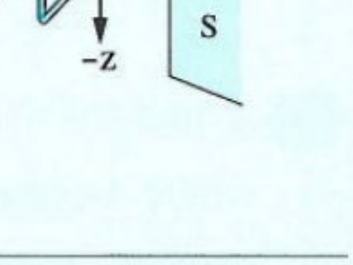


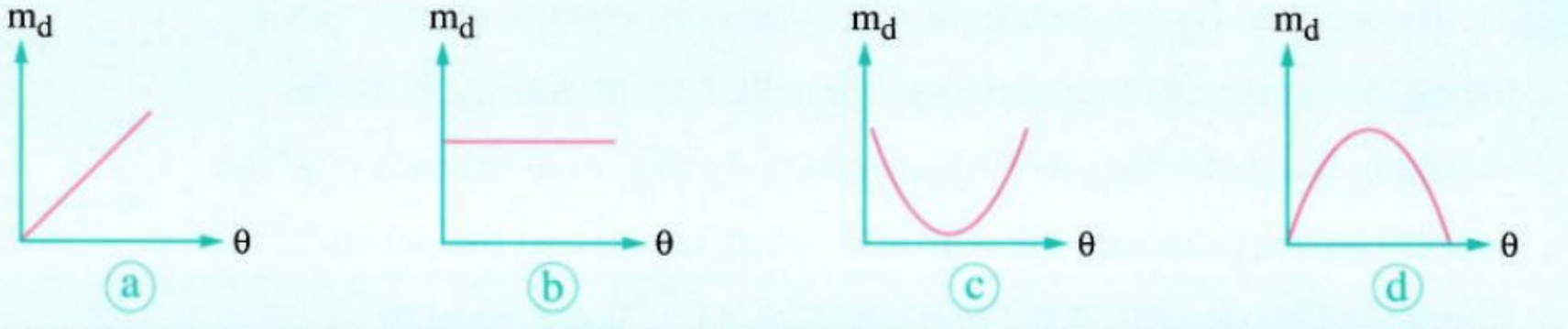
(a) Ejecting the coil from the field area in the direction of axis y
(b) Rotating the coil by 180° around axis x
(c) Rotating the coil by 180° around axis y
(d) Rotating the coil by 90° around axis z

18 Each of the following figures represents a straight wire placed close to a solenoid and each of them carries an electric current, in which of the following figures, the net magnetic flux density at the center of the solenoid (point c) is rightwards parallel to the plane of the page?



19 The opposite graph represents the variation of the torque (τ) that affects a rectangular current-carrying coil while being placed in a magnetic field versus the angle of rotation of the coil (θ) from a definite initial position, so which of the following graphs represents the relation between the magnetic dipole moment (m_d) of the coil and the angle of rotation (θ) through the same time interval?

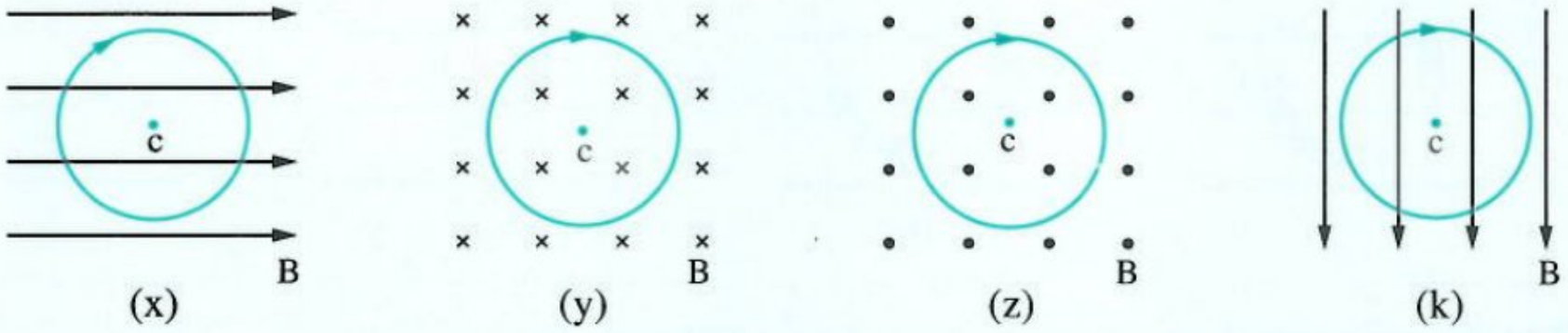




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20 An electric current passes through a metallic ring that produces a magnetic field of flux density B at the center of the ring (c), if the ring is placed in a magnetic field of flux density B and the position of the ring has been changed with respect to the direction of the field as in the cases x, y, z and k.



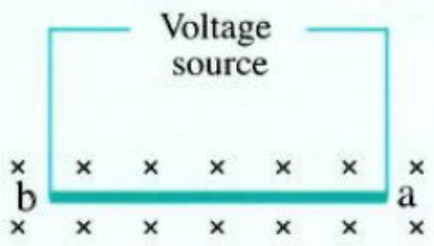
So, the correct order for the resultant magnetic flux density at the center of the ring in these cases .....

(a) $B_x > B_y > B_z > B_k$
(b) $B_k > B_x > B_y > B_z$
(c) $B_k = B_x > B_y > B_z$
(d) $B_y > B_k = B_x > B_z$

21 A straight wire of length 40 cm and resistance 8 Ω is coiled in the form of a square frame where a current of intensity I is passing through it and the potential difference across the ends of one of its sides equals 4 V. So, the magnetic torque that acts on the frame if it is placed parallel to a magnetic flux of density 0.1 T is equal to ..... .

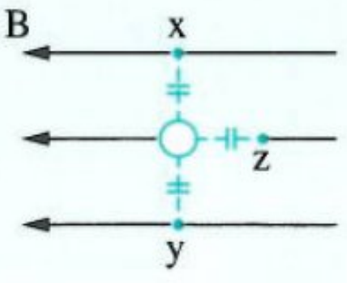
(a) 10^{-3} N.m
(b) 2×10^{-3} N.m
(c) 4×10^{-3} N.m
(d) 8×10^{-3} N.m

22 In the opposite figure, a horizontally placed wire ab of a mass per unit length 20 g/m is positioned perpendicular to a uniform magnetic flux of density 0.2 T which is directed into the page, hence the intensity and the direction of the current which if passed through the wire will generate a magnetic force causing the wire to be apparently weightless are ..... . (Take: $g = 10\ m/s^2$)



(a) 1 A, from a to b
(b) 2 A, from a to b
(c) 1 A, from b to a
(d) 2 A, from b to a

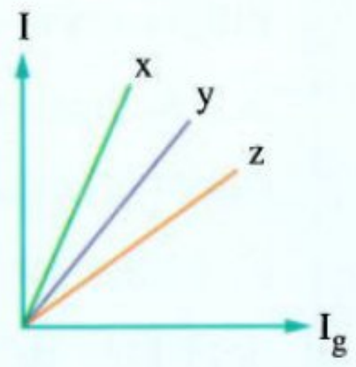
23 In the opposite figure, a straight wire carries an electric current while being perpendicular to a uniform magnetic flux of density B. If the resultant magnetic flux density at point y = zero, so the direction of the current passing through the wire and the ratio between the values of the magnetic flux density at the two points x, z $\left(\frac{B_x}{B_z}\right)$ are equal to ..... .



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	The direction of the current through the wire	$\frac{B_x}{B_z}$
(a)	outward	$\frac{2}{1}$
(b)	outward	$\frac{\sqrt{2}}{1}$
(c)	inward	$\frac{2}{1}$
(d)	inward	$\frac{\sqrt{2}}{1}$

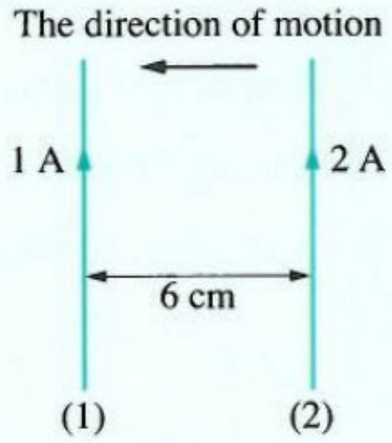
24 A multi-scale ammeter consists of a galvanometer and three resistances x, y and z, one of which can be connected in parallel as a shunt resistance, the opposite graph represents the relation between the reading of ammeter (I) and the current intensity (I_g) that passes through the galvanometer in the three cases.



In which of the three cases does the ammeter have greater sensitivity?

(a) In case of shunt x
(b) In case of shunt y
(c) In case of shunt z
(d) The three cases have the same sensitivity.

25 Two straight parallel wires (1), (2) carry electric currents of intensities 1 A, 2 A respectively as shown in the opposite figure, if wire (2) is moved a distance 2 cm towards wire (1), then what is the magnitude and the direction of the displacement of the neutral point?

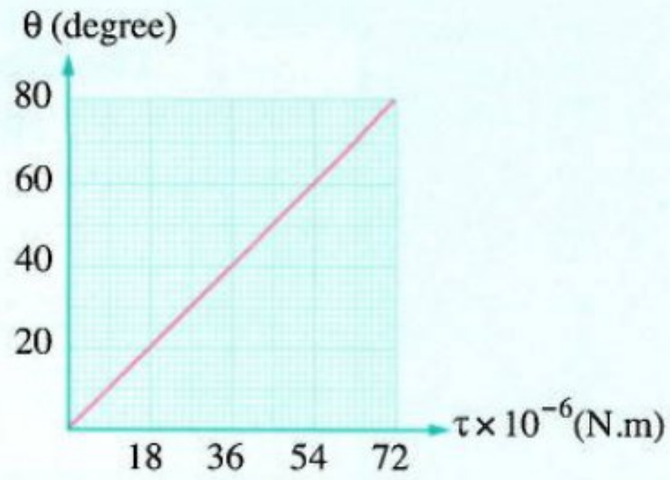


	The value of the displacement	The direction of the displacement
(a)	$\frac{2}{3}$ cm	towards wire (1)
(b)	$\frac{2}{3}$ cm	towards wire (2)
(c)	$\frac{4}{3}$ cm	towards wire (1)
(d)	$\frac{4}{3}$ cm	towards wire (2)

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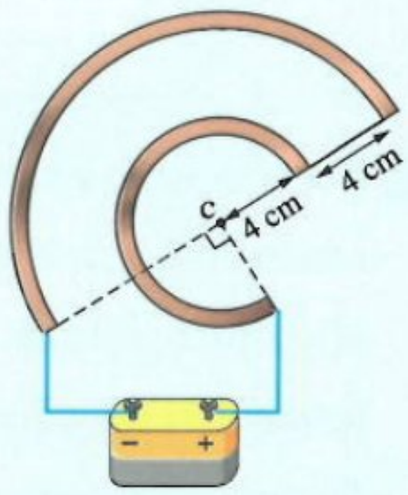
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26 A coil of galvanometer consists of 1200 turns, its face area is 15 cm^2 and it rotates in a uniform magnetic field of flux density 0.01 T, the opposite graph represents the relation between the deflection angle (θ) of the galvanometer's pointer from the zero position and the magnetic torque (τ) acting on the coil, then the sensitivity of the galvanometer equals ..... .



(a) 20 deg/mA (b) 50 deg/mA (c) 30 deg/mA (d) 60 deg/mA

27 Two parts of two circular metallic rings are connected in series to a battery as shown in the opposite figure, if the current intensity that passes through the battery is 2 A, then the magnetic flux density at the common center (c) equals ..... .



(a) 1.42×10^{-5} T (b) 1.57×10^{-5} T
(c) 3.14×10^{-5} T (d) 3.93×10^{-5} T

28 A uniform metallic wire of radius r_1 is divided into two equal parts (1), (2), part (1) is wrapped in the form of a solenoid whose turns are tangent to each other, while part (2) is wrapped in the form of a circular coil of center (c) of number of turns 100 turns and radius r_2 around the middle of the solenoid so that their axes, that are perpendicular to their respective cross-sections, coincide with each other. When a current of intensity I is passed through each of the two coils, the magnetic flux density at the center of the circular coil vanishes, so the ratio of $\left(\frac{r_1}{r_2}\right)$ equals ..... .

(a) 1 (b) $\frac{1}{10}$ (c) $\frac{1}{50}$ (d) $\frac{1}{100}$

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Second Answer the following questions

29 Figure (1) shows a circular metallic ring of radius 1 cm that carries an electric current of intensity I_1 where the ring is tangent to an insulated straight wire ab that carries a DC changeable electric current (I_2) and is placed in the plane of the ring. The graph in figure (2) represents the relation between the net magnetic flux density (B_c) at the center of the ring (c) due to the wire ab and the ring and the electric current intensity that passes in the straight wire (I_2),

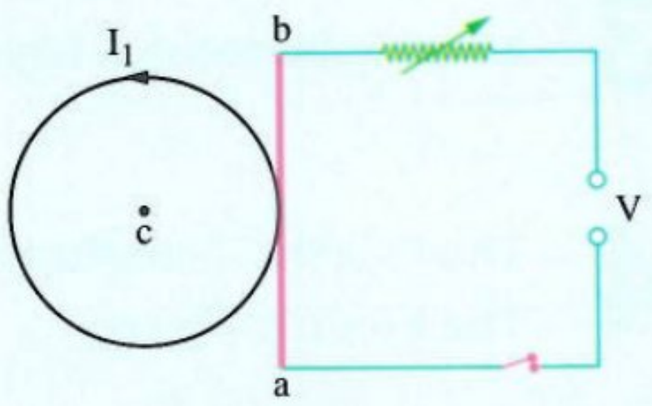


Figure (1)

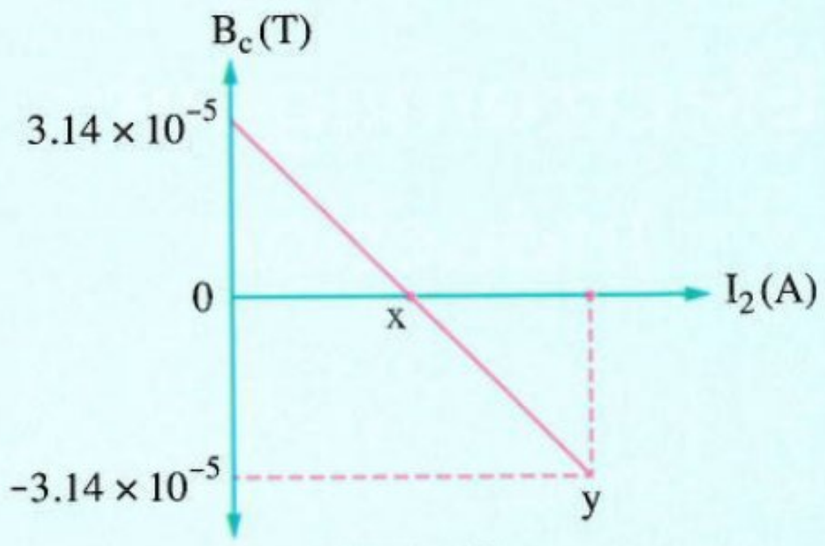


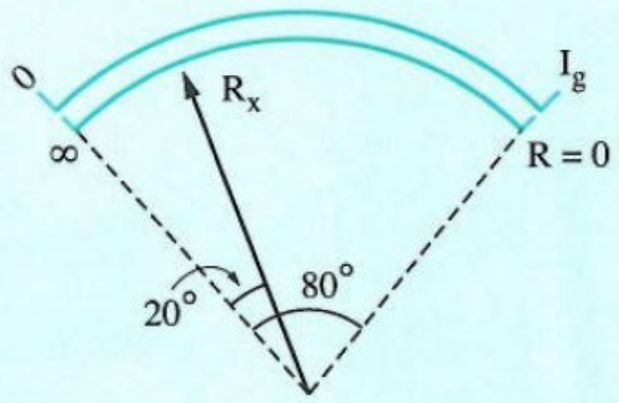
Figure (2)

(a) Find the value of the current intensity (I_1) that passes through the metallic ring.

(b) Find the value of the magnetic flux density produced due to the current (I_2) of the wire at each of the two points (x), (y) on the graph. (Where: $\pi = 3.14$)

.....

30 The opposite figure represents the scale of an ohmmeter of resistance 2000 Ω whose pointer has maximum deflection range of angle 80° from the zero position. When connecting the ohmmeter terminals to a resistor R_x, a current of intensity 200 μA flows through it and its pointer deflects as shown in the figure, **calculate:**



(a) The value of resistance R_x

(b) The electromotive force (V_B) of the battery used in the ohmmeter.

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